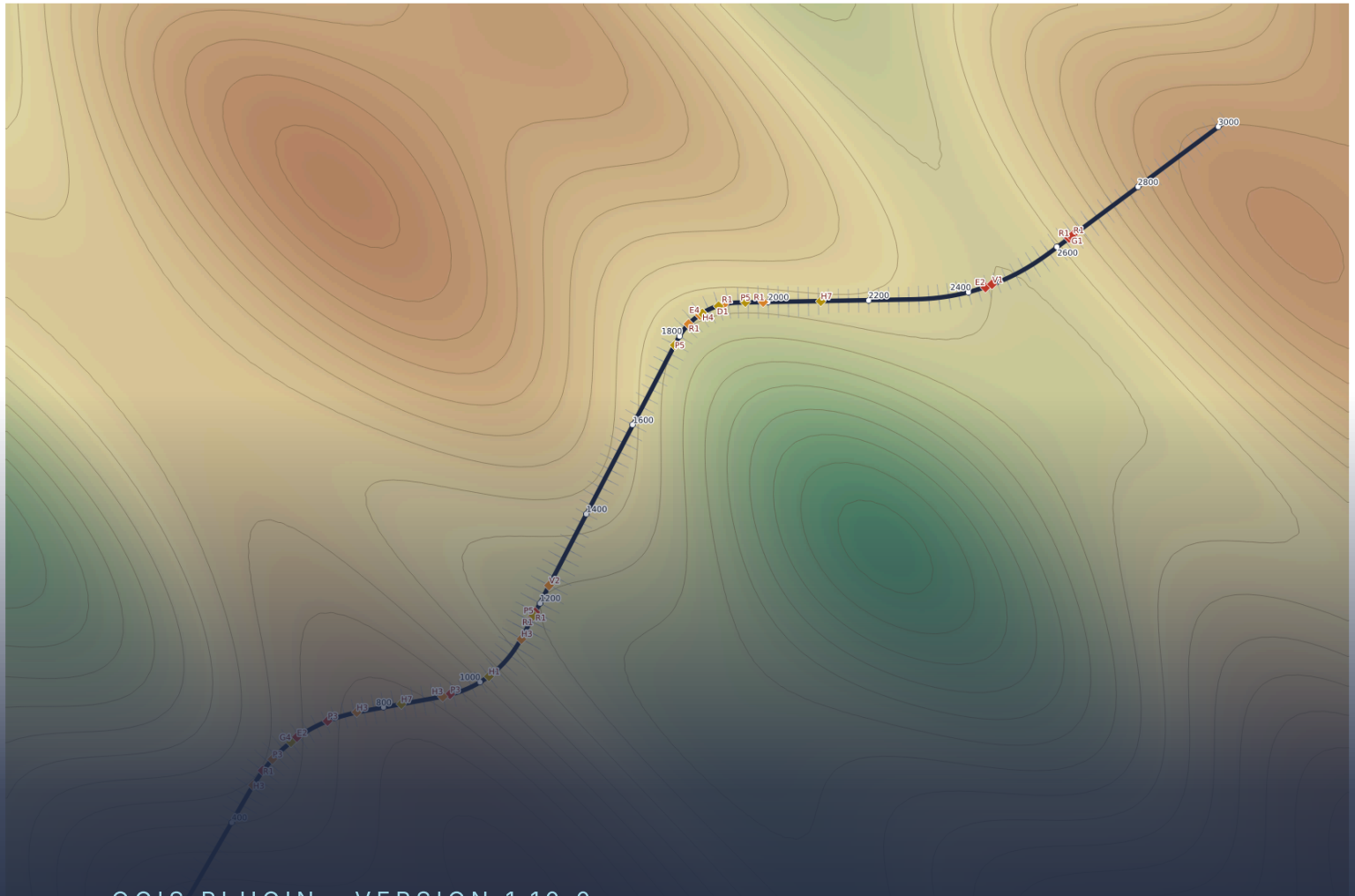


QGIS PLUGIN · VERSION 1.10.0

LandXML Road & Terrain GIS Tools

User Guide and Technical Manual

Author	Edmond Akello
Contact	akelloedmond@gmail.com
ORCID	0009-0009-5901-4574

Repository	github.com/EdmondAkello/LandXML-Road-and-Terrain-GIS-Tools
QGIS	3.44 to 4.x (Qt5 and Qt6 builds)
DOI	10.5281/zenodo.23402501 (Zenodo, all versions)
Cite as	Akello, E. and Winstead, C. (2026). *LandXML Road & Terrain GIS Tools* (version 1.10.0) [Software]. Zenodo. https://doi.org/10.5281/zenodo.23402501
Licence	GNU General Public License v2 or later
Edition	November 2026

All figures in this guide were produced by the plugin from a synthetic demonstration road with invented names, coordinates and terrain. No project data is shown.

Contents

ABOUT THIS GUIDE

0 What is new since 1.4

PART A — GETTING STARTED

1 Overview
2 Installing and updating
3 Preparing the LandXML export
4 Coordinates, axis order and units
5 Inspect LandXML first
6 Choosing surfaces, alignments and profiles

PART B — CONVERTING DATA TO GIS

7 Complete Road Design to GeoPackage
8 Terrain: TIN to GeoTIFF and boundaries
9 Alignments, stations and 3D centrelines
10 Profiles, labels and map overlays
11 Cross-sections, breaklines and feature lines

PART C — VIEWING, QUANTITIES AND REPORTS

12 Profile Viewer
13 Profile Cut/Fill Analysis
14 Corridor Section Quantities
15 Surface Cut/Fill and Grading Quantities
16 Cross-Section Sheets and Report
17 Road Design Report

PART D — GEOMETRIC DESIGN REVIEW

18 What the review does
19 Running a review
20 Design speed and speed changes
21 Checks and finding codes
22 Alignment phasing and recommended PVIs
23 Safety barriers and sight on curves
24 Taking recommendations back to Civil 3D
25 Design standards and custom criteria

PART E — TECHNICAL REFERENCE

26 Computation methods
27 Output layers and fields
28 Supported LandXML content and limits
29 Validation
30 Quality-control checklist
31 Troubleshooting
A Criteria tables

0 What is new since 1.4

The 1.4 guide covered conversion of LandXML to GIS layers. Versions 1.5 to 1.8 turn the plugin into a review and reporting tool for road designs:

1.5 · QGIS 4 and inspection
QGIS 4 support, Inspect LandXML, explicit units and CRS handling, OpenRoads Designer terrain and breaklines, profile labels and map overlays.

1.6 · Viewer and earthworks
Profile Viewer dock (click a road to see existing vs design), profile, corridor and TIN cut/fill quantities, mass haul and side-slope areas for grassing.

1.7 · Reporting
Name dropdowns and automatic surface choice, cross-section sheets (PDF/HTML/PNG), cross-sections in the viewer, and the Road Design Report.

1.8 · Design review
Geometric Design Review against Kenya RDM 1.3 (2025) or AASHTO 2018: phasing, sight distance, curves, grades, superelevation, design speed changes and barrier warrants.

1.8 · Back to Civil 3D
Recommended PVIs and curve lengths written as a Civil 3D "Create Profile from File" text file and as LandXML.

Fixes
Correct placement of station-relative cross-sections, hidden TIN triangles excluded, true arc-length stationing on curves and spirals.

PART A

Getting started

Install the plugin, export a suitable LandXML file from the design software, decide how its coordinates map to your GIS, and check the file before converting anything.

1 Overview

LandXML Road & Terrain GIS Tools is a QGIS Processing provider with a dockable profile viewer. It reads LandXML 1.0–1.2 exported by Autodesk Civil 3D, Bentley OpenRoads Designer and other design packages, and turns terrain and road design data into GIS layers, quantities, drawings and design-review reports. It needs no Python packages beyond those QGIS ships (NumPy, GDAL/OGR), and it never edits the source file.

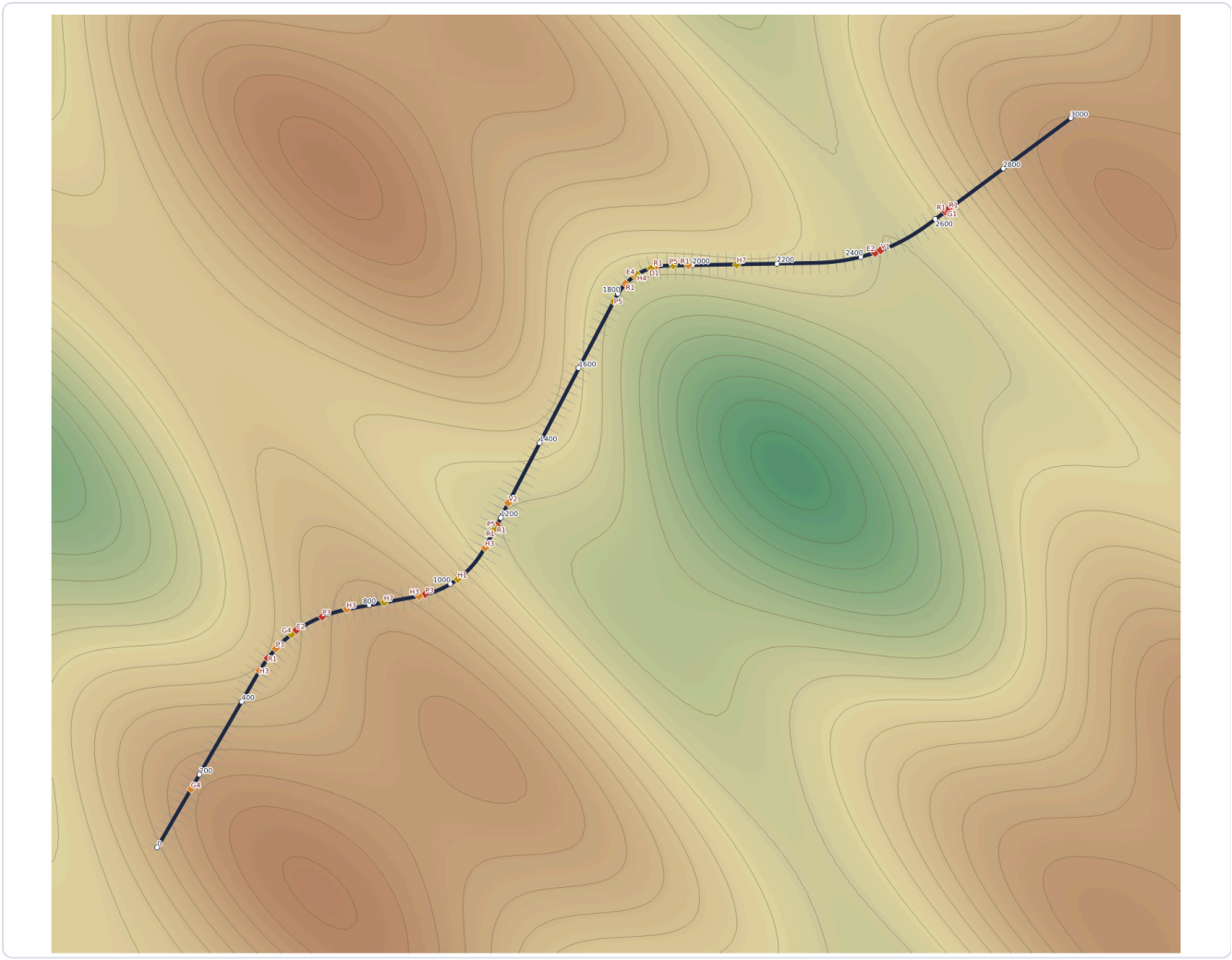


Figure 1. The demonstration road in QGIS: DEM and 5 m contours from the TIN, alignment and 200 m stations from Complete Road Design, corridor section lines, and design-review findings coloured by severity (red high, orange medium, yellow low) and labelled with their codes.

Toolbox group	Tools
1. Inspect and prepare	Inspect LandXML · Prepare LandXML from Geometry and Terrains
2. Road design extraction	Extract LandXML Profiles · Create 3D Road Centerlines · Extract LandXML Cross-Sections · Create Alignment Station Points · Export Complete Road Design to GeoPackage
3. Vector extraction	Extract LandXML Alignments to Lines
4. Terrain conversion	LandXML TIN to GeoTIFF
5. Surface extraction	Extract TIN Surface Boundary · Extract LandXML Feature Lines · Extract LandXML Breaklines
6. Earthworks and quantities	Profile Cut/Fill Analysis · Corridor Section Quantities · Surface Cut/Fill (TIN Difference)
7. Design review	Geometric Design Review (Phasing, Sight Distance, Standards)
8. Reports and sheets	Cross-Section Sheets and Report · Road Design Report
9. Site grading (land development)	Grading Quantities (Volume Surface) · Platform Levels and Falls · Basin Stage–Storage · Earthwork Balance (Volume Surfaces) · Grading Revision Comparison (Volume Surfaces)

Toolbox group	Tools
Outside the Toolbox	LandXML Profile Viewer dock and the **Pick Alignment Profile on Map** button (menu **Plugins → LandXML Road & Terrain**)

A typical workflow

1. **Export** the LandXML from the design software with surfaces, alignments, profiles, sample-line cross-sections and design speeds (section 3).
2. **Inspect** the file to see what it contains and to settle units and axis order (section 5).
3. **Convert** with Complete Road Design to get a GeoPackage, DEM and contours, and check placement against survey control (section 7).
4. **View and quantify:** open the Profile Viewer on the road, run the earthworks tools and plot the cross-sections (Part C).
5. **Review** the geometry against your design standard and take the recommended changes back to the design software (Part D).
6. **Report:** combine the Road Design Report, review report, section sheets and maps into the project report.

2 Installing and updating

From the QGIS plugin repository

1. Open **Plugins → Manage and Install Plugins...**
2. Search for **LandXML Road & Terrain GIS Tools** and press **Install** (or **Upgrade**).
3. Open **Processing → Toolbox** and expand **LandXML Road & Terrain GIS Tools**. The Profile Viewer button appears on the Plugins toolbar.

From a ZIP file

Use **Plugins → Manage and Install Plugins → Install from ZIP**. The ZIP contains one folder, `landxml_tin_to_geotiff`. This is the plugin's permanent folder name, kept since 1.4 so that new versions replace old ones; installing a copy under a different folder name gives two plugins side by side.

> **REQUIREMENTS**
> QGIS 3.44 or later, including QGIS 4 (Qt6). Metric and imperial files are both read; the design review and earthworks reports assume metric data.

3 Preparing the LandXML export

The plugin can only use what the export contains. Decide what you need before exporting, and keep one export per review so results can be traced.

To use	Include in the export
Terrain, DEM, contours, TIN difference	Surfaces (TIN). Civil 3D marks triangles hidden by a surface boundary as invisible; the plugin excludes them.

To use	Include in the export
Alignments, stations, 3D centrelines	Alignments with their geometry (lines, curves, clothoid spirals).
Profiles, Profile Viewer, cut/fill	The design profile (layout profile) and the existing-ground surface profile of each alignment.
Corridor quantities, section sheets, barrier and sight checks	Sample-line cross-sections with the existing-ground and corridor surfaces sampled, and the corridor shapes and links (pavement layers, Datum). Enable cross-section export in the LandXML export settings.
Design review by speed	Design speeds: in Civil 3D set them in **Alignment Properties → Design Criteria** (speed stations) *before* exporting. They are written as `SpeedStation` features.
Superelevation diagram and checks	Alignment superelevation (Civil 3D writes transition stations and full rates).

In Civil 3D the command is **Output → Export to LandXML**; in OpenRoads Designer use the LandXML export. The plugin does not read DWG or DGN files.

> **TIP**
> Large corridor exports are fine: the plugin has been run on 75 MB files with thousands of cross-sections. Name dropdowns scan a file in under half a second.

4 Coordinates, axis order and units

Coordinate handling is explicit in every tool that writes map geometry. The plugin never guesses a CRS or transformation; a CRS declared in the LandXML is shown for reference only.

Coordinate interpretation	Use when
Use stored coordinates	The stored values are already easting, northing in the output CRS.
Swap X / Y	The file stores Northing, Easting (Y, X), the standard LandXML plan-coordinate order normally used by **OpenRoads and Civil 3D**. This produces GIS X = Easting, Y = Northing.
Apply X / Y offset	A known translation (for example a false origin) must be added.
Swap X / Y + offset	Both of the above.
2-D Helmert (similarity)	A known translation, rotation and scale relate the design grid to the map grid.
Reproject from source CRS	The stored coordinates are valid in a known CRS that differs from the output CRS. You must select the source CRS.

Output CRS is always chosen by you. Use the same interpretation and CRS for every layer of a project, and check one feature against known survey control before converting a whole design. If you keep the stored order but the first coordinate looks like a northing (for example 9,886,035 with an easting-sized second value), every tool warns you; it never swaps on its own.

OpenRoads axis-order reminder. Every coordinate-interpretation selector includes a reminder and detailed help. Standard LandXML uses Northing, Easting, with elevation third (see the ICSM mapping); the downloaded FHWA OpenRoads examples follow this order. Verify a known control point instead of assuming every version

or edited export behaves identically. Use stored coordinates if the file already contains Easting, Northing. Preparation preserves axis order; apply the same choice to matching geometry and terrains. Profile station/elevation and section offset/elevation pairs are not plan coordinates and must not be swapped.

Units. Metres, international feet and US survey feet are kept distinct. When a file declares a horizontal unit, the output CRS must use the same unit or the tool stops. Vertical units are never inferred from horizontal ones; if the file does not declare them, the log says so.

5 Inspect LandXML first

Inspect LandXML (Utilities) reads the file without converting anything and reports: the producing software and version, horizontal and vertical units, declared CRS, an axis-order warning, surfaces with point and face counts, alignments, design and surface profiles, cross-sections, breaklines, feature lines, coordinate bounds, the elevation range and any warnings. It can save the report as JSON for the project record.

Use it to confirm that the file holds what you expect (for example that cross-sections are present before running corridor tools) and to decide on the coordinate interpretation.

6 Choosing surfaces, alignments and profiles

Remembering project settings. After a successful run from a Processing dialog, the tool remembers its input files, selections, units, intervals, coordinate settings and design criteria in the current QGIS project. Save the project to retain them after closing QGIS. Each tool keeps separate settings; a new project starts with defaults. Explicit inputs from Processing history override saved inputs. Output destinations and the Prepare LandXML coordinate/datum confirmation are not remembered. Failed or cancelled runs leave the saved settings unchanged. Scripts, models and batch runs keep their normal parameter behavior.

Every name field is an editable dropdown filled from the chosen file and refreshed when the file changes. You can also type: matching ignores case and accepts a unique part of a name, so `eg` finds `EG` and `ogl` finds `OGL-LOT_REV 0`. An unknown name is reported with the list of names available.

Leave surface names blank to let the tool decide from evidence in the file; the choice is written to the log:

- Existing ground is the surface named inside an existing-ground `ProfSurf`.
- The design surface is a corridor surface listed by a `Roadway`, preferring the datum (bottom) surface for earthworks.
- Surface Cut/Fill first uses a Civil 3D volume-surface pair (`SurfVolume`) if the file records one.

PART B

Converting data to GIS

Turn terrain, alignments, profiles and sections into GeoPackage, GeoTIFF and 3D vector layers for mapping and analysis.

7 Complete Road Design to GeoPackage

For a first import, run **Export Complete Road Design to GeoPackage**. In one pass it writes an engineering GeoPackage and, optionally, a DEM and contours.

Automatic output selection. Leave the Include options enabled to export all available data. Outputs with no exportable features are skipped after applying the alignment/surface filters and checking dependencies. Missing profiles, sections or terrain do not need to be unchecked first. Empty layers are omitted from the saved package and from QGIS; the Processing log reports skipped outputs. Invalid-data errors and warnings still apply. If no engineering vector layers are produced, no engineering GeoPackage is retained; DEMs can still be exported independently. A contour GeoPackage is omitted when the requested interval produces no contour lines.

Automatic terrain names. DEM and contour filenames retain the XML basename, add the embedded surface name, and include a supported role such as existing ground or finished grade. For example, surface `FG Road` in `combined.xml` produces `combined_Finished_Grade_FG_Road_DEM.tif` and `combined_Finished_Grade_FG_Road_Contours.gpkg`. Roles come from surface names/descriptions, explicit state or Prepare LandXML metadata; datum and generic design/proposed surfaces are not labelled finished grade. Unknown or conflicting roles retain a neutral name. Unsafe characters are replaced, unnamed surfaces get numbered fallbacks, and duplicate filenames receive numeric suffixes. The mixed engineering package remains `combined_GIS.gpkg` with its existing tables. Other tools keep user-selected output paths. Earlier exports are not renamed; update scripts that expect the former numbered terrain filenames.

Parameter	Default	Guidance
Coordinate interpretation, Output CRS	—	As verified in section 4.
Alignment / TIN surface	blank = all	Pick from the dropdowns to limit the export.
DEM resolution	5	Pixel size in map units. Finer cells do not add detail missing from the TIN.
Contour interval	5	0 skips contours.
Alignment segment length	5	Densification of curves and spirals.
Station interval	20	Station points at whole multiples of the interval.
Profile interval	5	Sampling of profile graph lines.

Parameter	Default	Guidance
Include ... toggles	on	Export all available data automatically; missing outputs are skipped. Turn off an option to exclude available data you do not need.
Load exported layers into the project	on	After success, QGIS loads the generated GeoPackage layers, retained DEMs and contours into the run's project. Turn off to export files only. Temporary DEMs used only for contours and disabled outputs are not loaded. Failed/cancelled exports do not queue partial results. Runs without a project still export normally. Source XML links support the Profile Viewer. Profile graphs remain station/elevation diagrams without a map CRS.

Layer	Contents
`alignments`	One line per alignment with length and start/end stations.
`alignment_elements`	Each line, curve and spiral with direction, radius, curve and spiral length.
`centerlines_3d`	3D centreline over the range covered by the design profile.
`profiles`, `profile_controls`	Profile graph lines and VPI/VPC/VPT controls in station/elevation graph coordinates (no map CRS).
`station_points`	Points with station and bearing.
`cross_sections`	Section lines (3D) for every sampled surface, placed along the alignment.
`feature_lines`, `breaklines`	Design linework with source names, codes and IDs.
`surface_boundary`	Outer edges of each TIN.
`*_DEM.tif`, `*_Contours.gpkg`	Float32 DEM and contour lines (optional).

EXISTING FILES

The tool refuses to overwrite an existing GeoPackage. Choose a new output folder for each run.

Station and contour label conventions. Station-point outputs include `station_label`, using declared source units: foot-based stationing displays `50+00`, `7+00` or `700+00`; metric follows the existing formatter (metres: `5+000`). Whole stations omit `.00`; fractional stations retain two decimals. Unknown units keep plain numeric values. Contour labels omit trailing decimal zeros but preserve fractional elevations. Identified existing-ground minor contours are brown/dashed; proposed/design minor contours are blue/solid, with thicker minor lines for visibility. Datum is purple and unknown surfaces are green. Numeric stations, elevations and geometry are unchanged.

Default QGIS styles. Complete-package layers and individual vector extractions receive consistent road-review presets: blue alignments/centerlines, small dark station markers, orange profile controls, purple cross-sections, grey boundaries, brown breaklines and green feature lines. Surface/profile attributes can distinguish brown existing ground, blue design/finished surfaces and purple datum/subgrade; unknown names retain the base color. Station labels use source station values; existing profile labels are preserved. Contours use thin minor lines and heavier, labelled index lines every fifth contour interval. Defaults are saved in the GeoPackage

and restored when reopening it in QGIS, including file-only exports. Other GIS applications may ignore them. You can edit and save your own default styles afterwards. Memory outputs are styled for the current run; other formats do not get style sidecars. Existing earthworks styling is preserved; DEM styling is outside this first version. Styling changes presentation only, with no geometry, unit or quantity changes. A style-save failure is reported as a warning without discarding the data.

Prepared LandXML files retain original surface names and source filenames in preparation metadata. Automatic terrain names use that original surface name instead of the generated `Prepared …` name when available. The source filename also contributes role evidence, allowing a proposed terrain from a named finished-surface export to be labelled finished grade. Datum/subgrade remains distinct, and role wording already present in the surface name is not repeated.

8 Terrain: TIN to GeoTIFF and boundaries

Prepare separate geometry and terrain exports

Prepare LandXML from Geometry and Terrains (Utilities) accepts a primary alignment/profile LandXML and optional existing-ground and proposed terrain files. Select a surface when a supplied terrain file has several. For an embedded terrain, leave its file blank and select its surface. All inputs must already share coordinates, units and vertical datum: conflicting declarations are rejected, missing units require explicit selections, and missing metadata requires confirmation. No reprojection, axis swapping or unit conversion is performed.

Save to a new LandXML, then use it in the existing viewer, reports and analysis tools. Original design profiles and existing section records remain intact; the tool fills missing terrain profiles and section surfaces. A complete input needs no separate terrain files. Defaults are 5 source units between profile samples, 20 between sections and 50 left/right widths; adjust them to suit the project.

Samples use the original visible TIN triangles. Terrain-edge crossings are retained; hidden triangles are excluded. Profile gaps remain disconnected, and disconnected section surfaces are omitted with warnings. Continuous partial sections stop at the terrain boundary. Missing section stations interrupt volume integration, but gaps between sampled stations still require inspection and exclusion. Generated sections contain terrain surfaces, not pavement layers or reconstructed design PVIs. Repeated runs reuse generated records; conflicting generated names or changed source terrain are rejected. The output is written atomically and never replaces an input.

Large-grid warning. TIN to GeoTIFF, Complete Road Design and Surface Cut/Fill provide **Warn above grid size (million cells; 0 = disable)**, defaulting to 250 million cells. Larger grids log a warning with the cell count and minimum grid memory, then continue. Temporary arrays and surface comparisons need additional RAM. Increase the pixel size if memory is insufficient, or set the threshold to 0 to disable the warning.

LandXML TIN to GeoTIFF rasterizes one TIN surface by linear interpolation inside each triangle. Choose the surface, the pixel size and a NoData value outside the elevation range. Triangles flagged invisible (`i="1"`, written by Civil 3D for triangles hidden by a boundary) are left as NoData, so the raster follows the surface boundary instead of filling holes and outer concavities.

Extract TIN Surface Boundary writes the outer edges of the visible triangles as lines.

9 Alignments, stations and 3D centrelines

- **Extract LandXML Alignments to Lines** writes plan geometry. Lines, circular curves and explicitly declared clothoid spirals are supported; a segment that does not connect or does not match its declared length is

reported and skipped rather than forced into place.

- **Create Alignment Station Points** places points at whole multiples of the interval (20, 40, 60 …) even when the alignment starts at a fractional station. Stationing uses true arc length along curves and spirals.
- **Create 3D Road Centerlines** gives the centreline Z from the matching design profile. Where the profile covers only part of the alignment, only that part is written and the omitted range is reported; missing elevations are never set to zero.

10 Profiles, labels and map overlays

Extract LandXML Profiles writes profile graph lines and control points (VPI from the file, VPC and VPT computed for each parabolic curve). Labels show the station (`0+000.00` for metric files), elevation, incoming and outgoing grades and K. A **vertical exaggeration** scales the plotted Y while the `elevation` attribute keeps the source value.

To see the profile next to the road in the map, add the **Map profile lines** output with the same coordinate interpretation and CRS as the alignment. The profile is drawn beside the alignment at the chosen offset (positive left, negative right). These are schematic graphics, not surveyed features.

K is the curve length divided by the algebraic difference in grade in percent. Flat curves have no finite K.

11 Cross-sections, breaklines and feature lines

Extract LandXML Cross-Sections handles both forms found in LandXML:

- **Absolute** sections (`PntList3D`) keep their coordinates.
- **Station-relative** sections (`CrossSectSurf/PntList2D` offset–elevation pairs, as Civil 3D writes for sample lines) are placed along their alignment as 3D lines, one per sampled surface. Versions before 1.6 placed these at false coordinates.

Extract LandXML Breaklines and **Extract LandXML Feature Lines** keep the design linework with its source names, descriptions, codes and IDs. OpenRoads breaklines carry codes and IDs but no names; the plugin keeps those without treating IDs as names.

PART C

Viewing, quantities and reports

Look at the design the way a reviewer does: long-sections and cross-sections on demand, earthworks quantities and report-ready drawings.

Saving reports as PDF

Every HTML report is laid out for A4 printing. Open it in Chrome or Edge, choose **Print → Save as PDF** and untick *Headers and footers*. The report prints its own title and page numbers, keeps chart colours and legends, keeps each heading with its chart, repeats table headers on every page and starts each alignment on a new page. The Road Design Report opens with a table of the alignments it contains.

12 Profile Viewer

> **WHERE TO FIND IT**
>
> The Profile Viewer is a panel, not a Processing tool, so it is *not* listed in the Processing Toolbox. Open it from **Plugins → LandXML Road & Terrain → LandXML Profile Viewer** or **Pick Alignment Profile on Map**, or press the plugin's button on the Plugins toolbar (the same icon as the plugin). The panel docks along the bottom of the QGIS window; drag its title bar to move or float it. Once opened, it is also listed under **View → Panels**.

Press **Pick Alignment Profile on Map** and click an alignment, or any layer with an `alignment_name` or `alignment` field (station points, cut/fill segments, cross-sections). The dock opens that alignment's existing-ground and design long-section with the clicked chainage marked.

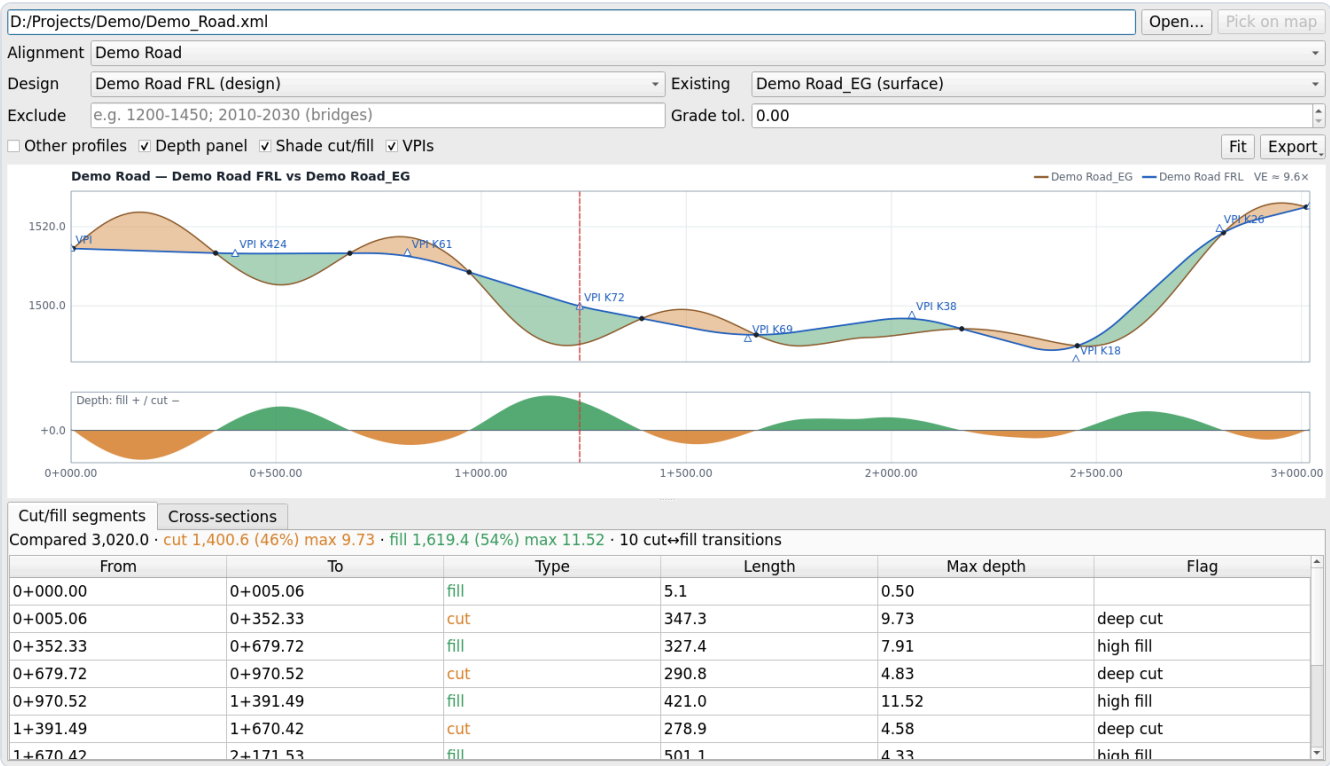


Figure 2. Profile Viewer: existing ground (brown) against design (blue) with fill shaded green and cut orange, VPIs labelled with K, cut↔fill transitions (black dots), the depth panel and the cut/fill segment table with high-fill and deep-cut flags.

- **Profiles:** the longest existing-ground profile and the first design profile are used by default; any profile can take either role, and **Other profiles** overlays the rest.
- **Interaction:** hover for chainage, both levels and depth (the map marker follows); wheel to zoom, drag to pan, double-click to fit, click to centre the map.
- **Grade tol.** treats small depths as on grade so ground noise does not create transitions. **Exclude** removes structures such as bridges, e.g. `1200-1450; 2010-2030`.
- **Export:** PNG, SVG, PDF (A3), clipboard image, station table CSV, segment CSV and an HTML report.

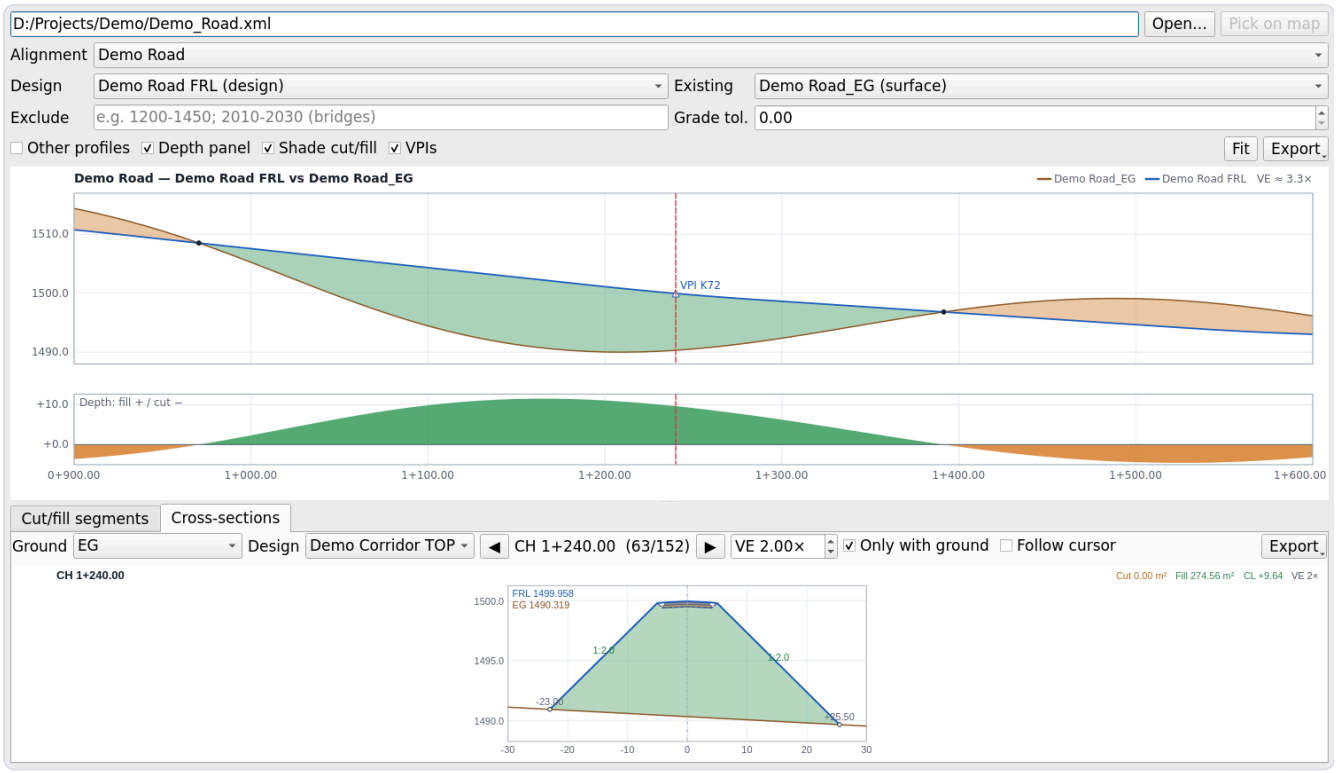


Figure 3. The Cross-sections tab follows the profile marker: here an embankment at CH 1+240 with 1:2 slopes, the pavement layers, daylight offsets, FRL and EG levels, and the fill area. ◀ ▶ step through sections; vertical exaggeration and export are on the same row.

Layers created by the plugin remember their LandXML file in the QGIS project, so clicking them opens the right file. For other layers the viewer asks for the file once.

Interactive Station & Offset Locator

Open **Plugins → LandXML Road & Terrain → Station & Offset Locator**. This separate dock can be moved or floated in QGIS 3.44 and QGIS 4. Open the imported LandXML file and select its alignment. Explicitly choose the same **Coordinate interpretation (OpenRoads: usually Swap X / Y)**, output CRS, shifts, rotation, scale and source CRS used for import; these are not inferred from the layer or map. Verify a known control point.

1. Press **Click map** and click anywhere. The dock displays station, signed perpendicular offset and nearest distance. Negative offsets are left and positive offsets are right, facing increasing station.
2. The red circle marks the corresponding alignment point; the blue cross marks the station/offset position. Beyond endpoints, station is clamped and nearest distance includes the distance beyond the endpoint; the perpendicular offset does not.
3. Enter a numeric station or engineering notation with a plus separator, and a signed offset, then press **Locate station / offset**. For feet, `803+19.359` means `80319.359`; for metric source units, `1+025.359` means `1025.359`. A leading minus applies to the whole station. Engineering notation requires declared horizontal units; otherwise enter a numeric station. Out-of-range stations and non-finite inputs are rejected.
4. The station field and result always display **engineering notation with three decimal places** after map clicks and manual lookup, including when a plain number was entered. Clicked offsets also show three decimals. Without declared horizontal units, stations stay numeric. Marker coordinates retain calculation precision.
5. Choose another map tool or close the dock to stop picking. Closing hides the markers; changing alignment or coordinate settings invalidates old results. Clearing the project clears the loaded file.

Stations and offsets use source horizontal units, with no implicit conversion between metres, feet or US survey feet. Helmert scaling affects map geometry but preserves source station/offset values. With reprojection, calculations use the projected source CRS and markers are transformed into the map CRS. Geographic calculation CRSs are rejected. Undeclared units are labelled; elevations and vertical datums are not changed.

Tangents, circular curves and declared clothoids use the shared stationing helper with true source distances, so chord lengths do not accumulate into stationing error. Position and direction are interpolated on sampled geometry. **Maximum curve segment** defaults to 0.1 source units; reduce it for tight curves or large offsets. Accuracy depends on this sampling and source geometry. Missing start stations, invalid station equations, duplicate names, disconnected geometry and unsupported spiral types are reported rather than guessed.

Station equations

Declared roadway equations support increasing station regions with ahead/back jumps. The locator uses engineering station input: metric `1+950.000`, feet `19+50.000`. If a back jump repeats a station, choose its **Station region (equations)**; Automatic requires one physical position. Forward-jump gaps return “station does not exist”. Map clicks show region, internal station and distance along. At an equation both back/ahead labels identify the same point; the ahead region owns the boundary by default. Offsets keep the alignment travel direction.

Existing numeric station fields, profiles/sections, sampling intervals, quantity integration, exclusion ranges and design-speed ranges remain **continuous internal station** (alignment start plus distance). Station points always add `engineering_station`, `station_region` (1-based, 1 without equations), and `distance_along`; their label includes the engineering region. Complete Road Design includes a `station_equations` table. Other existing layer station labels remain internal. HTML reports explain this basis and list the equations; profile chart exports identify internal stations, and section drawings include engineering region and internal station. Profile Viewer CSVs retain existing columns and append engineering station, label, region and distance for numeric station columns.

Example: internal 2000, back 2000 = ahead 1900 makes engineering 1950 identify internal 1950 in R1 and 2050 in R2. A jump from 2000 back to 2100 ahead leaves engineering 2050 without a position. No road length or volume is added by either jump. International and US survey feet remain distinct. Inspection records invalid equations as `station_equation_error` with a warning; independent terrain exports and surface cut/fill remain usable. Station-dependent tools reject the equation set with a clear QGIS error. Complete Road Design rejects the package before creating outputs. Explicit decreasing progression is rejected. Missing back station is derived from the preceding region; omitted increment means increasing.

Prepared and recommended-profile LandXML retain equations and source internal geometry stations. Interpretation and validation notes distinguish Autodesk documentation, synthetic metadata variants and the private OpenRoads reference comparison. Nonstandard exporter station conventions require independent verification; railway posting and manual equation editing are not supported.

13 Profile Cut/Fill Analysis

Earthwork reporting units (all three quantity tools). Choose **Volume units**: Automatic (default), Cubic yards (CY), Cubic feet (CF), or Cubic metres (m³). Automatic uses CY for foot-based projects and m³ for metric projects. An explicit choice converts all cut, fill, net, material volumes and mass-haul ordinates into that unit, regardless of the input unit system. Report labels update to match; interval and footprint tables include `volume_unit`, and corridor/surface Processing results include `VOLUME_UNIT`. Numeric volume field names are unchanged, but values follow the selection: regenerate older reports and tables. Lengths, depths and areas retain their coordinate units; depth rasters are not converted to volume units.

US survey feet use their exact metre definition; CY uses the international yard and CF the international foot. Surface Cut/Fill uses the output CRS horizontal unit for Automatic and accounts separately for source elevation units. Recorded Civil 3D totals are converted from their declared volume unit into the selected output unit before comparison; missing or unsupported declarations omit that comparison. Volume estimates require known horizontal and vertical units; use **Prepare LandXML** to supply missing units explicitly. Profile/section quantities require matching horizontal and vertical units.

Profile Cut/Fill Analysis (Existing vs Design) compares the design profile with the existing-ground profile. Depth is design − ground (positive fill). Every breakpoint is kept and zero crossings are found exactly.

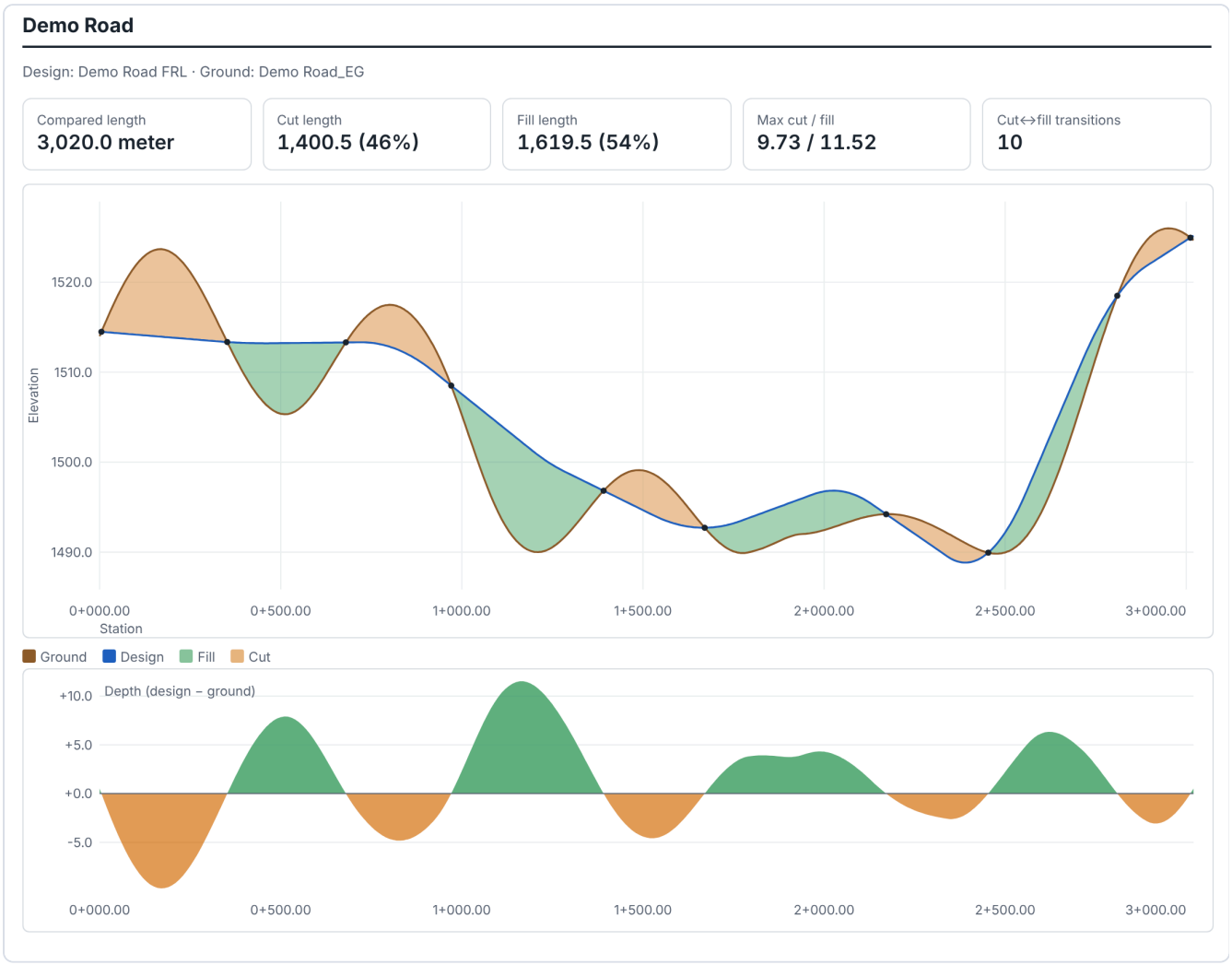


Figure 4. HTML report of Profile Cut/Fill Analysis: summary, shaded long-section and depth diagram.

Outputs: a station table at any interval, cut/fill segments with length, maximum and mean depth and *high fill / deep cut* flags (default 3 m), transition points on the map, and the report. A **formation width** adds a level-section estimate of volumes and side-slope areas for early design when no corridor sections exist; it ignores cross-fall, pavement depth and ditches.

14 Corridor Section Quantities

Corridor Section Quantities works from the station-relative sections exported from sample lines.

Demo Road

Ground: EG · Design: Demo Corridor TOP · Datum on · flat 1:10 · steep 1:1.5 · max spacing 100 · cut factor 1

Cut volume	Fill volume	Net (fill − factored cut)	Embankment slopes (grassing)	Cut slopes (grassing)
116,042	164,911	48,869	33,807	24,809

Steep slopes (stabilise)	Integrated length	Sections with earthworks
0	3,020.0	152 / 152

Corridor material quantities

Shape	Volume
Asphalt	1,057.00
Base	4,590.40
Subbase	4,952.80

Figure 5. Corridor quantities: cut and fill volumes by average end areas, net volume with the cut factor, embankment and cut slope areas for grassing, and volumes of each corridor shape.

- **Earthworks:** cut and fill areas between ground and design, average-end-area volumes and a mass-haul ordinate with an optional cut-to-fill factor. With **Use Datum** (default), earthworks are measured to the Datum links (bottom of pavement).
- **Materials:** volume of every closed corridor shape (asphalt, base, subbase...).
- **Side slopes for grassing:** sloped areas from the formation hinge to daylight, split into embankment and cut slopes. Segments flatter than 1:10 (shoulders, ditches, berms) are excluded; slopes steeper than 1:1.5 are reported separately as needing stabilisation beyond grassing.
- **Map layers:** side-slope band polygons per interval and side, and the earthworks footprint between daylight lines.

Mass haul ordinate

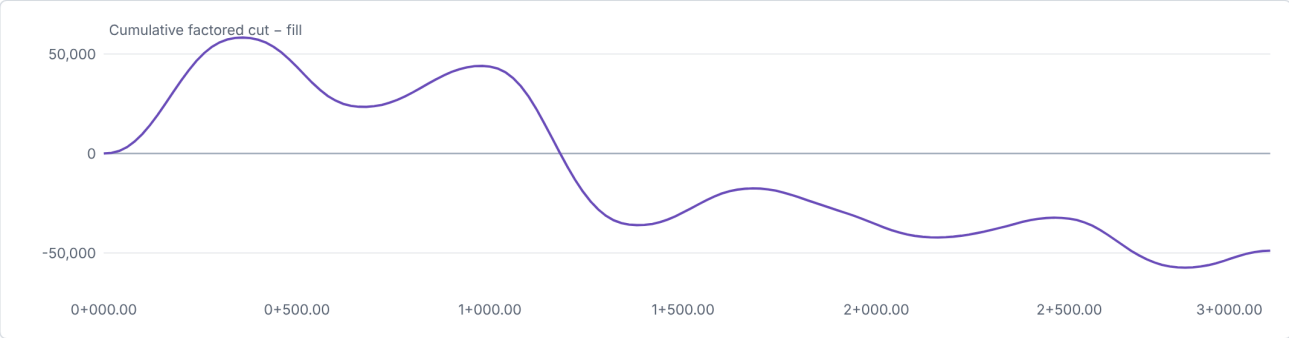


Figure 6. Mass-haul ordinate (cumulative factored cut − fill). Rising lengths are in surplus, falling lengths in deficit.

Sections farther apart than the maximum spacing (default 100) and excluded ranges are not integrated; the report lists what was skipped. No curvature correction is applied.

15 Surface Cut/Fill and Grading Quantities

Surface Cut/Fill (TIN difference)

Grids two TINs on a shared raster and writes compare − base as a GeoTIFF styled orange (cut) to green (fill), with cut, fill and net volumes. Leave both names blank to use a recorded Civil 3D volume-surface pair, or

existing ground against the corridor datum surface. When the file contains a `SurfVolume` for the same pair, the Civil 3D volumes are printed beside the grid result for comparison.

Grading Quantities (volume surface)

The site-grading tools are in their own toolbox group, **9. Site grading (land development)**, apart from the road-corridor earthworks. If the file does not declare its vertical unit (Civil 3D metric exports usually omit it), choose **Vertical unit** in the tool; it is never inferred, and a choice that contradicts a declared unit stops the run.

For site grading, export the Civil 3D **TIN volume surface** (design minus existing ground) with the other surfaces, and run **Grading Quantities (Volume Surface)** on it. Each triangle is integrated exactly as a linear prism and split at zero depth, so the result does not depend on a grid. The tool reports cut, fill and net volumes; plan, cut, fill and at-grade areas; maximum depths; and the exact plan area in each depth band (default limits −5, −3, −2, −1, −0.5, 0, 0.5, 1, 2, 3 and 5; a band includes its lower limit). Volumes are in place, with no shrinkage or bulking factors. Outputs are an HTML report and an optional depth-band CSV.

Without a volume surface. OpenRoads, and Civil 3D files exported without a volume surface, carry only the design and existing-ground surfaces. Enter the design surface and an **Existing-ground surface**: the tool builds the depth surface (design − existing) itself by clipping every design triangle against each existing triangle it overlaps. Both surfaces are linear on every piece, so the depth is linear too and the prism integration stays exact; the result is the TIN volume surface Civil 3D would build from the same two surfaces. Only the area both surfaces cover is counted, and any design area outside the existing ground is reported in the log. Earthwork Balance takes each site as `Design | Existing`, and Grading Revision Comparison takes an existing-ground surface for each revision.

Reconciliation. Enter the cut, fill and plan area from the Civil 3D cut/fill summary (unadjusted, factor 1.0) in the reported units. Each value is compared with the computed one and the run reports **PASS** when every difference is within the tolerance (default 0.01), otherwise **FAIL**. Civil 3D tables are rounded to two decimals, so differences up to 0.005 are rounding.

Leave the surface blank only when the file identifies exactly one volume surface (a Civil 3D volume record, or one surface named as earthworks, volume, cut/fill or difference); otherwise choose it. If the chosen surface has depths beyond 100 vertical units the log warns that it looks like an elevation surface. Civil 3D metric exports often omit the vertical unit; use **Prepare LandXML** to declare it.

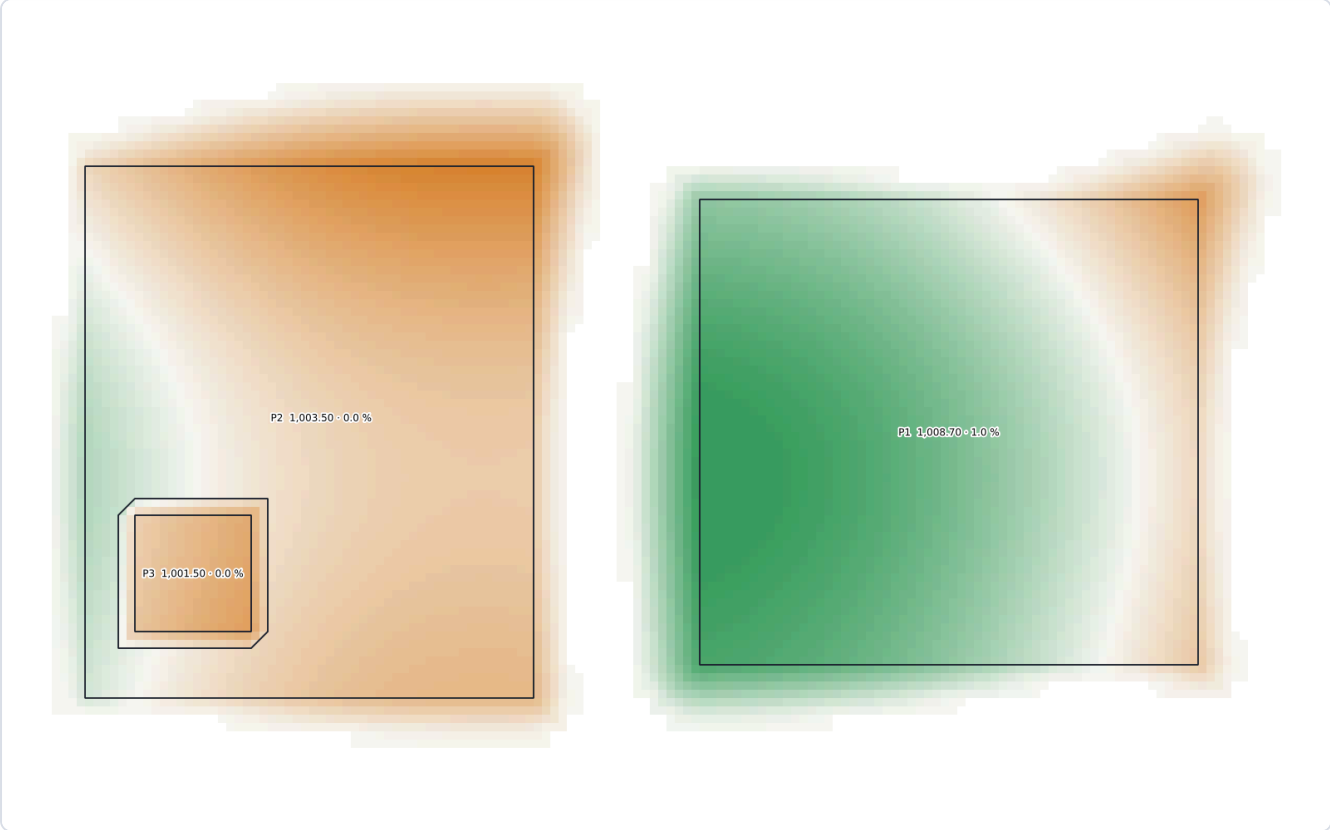


Figure 7. Synthetic grading demo: depth of the volume surface (cut orange, fill green, up to ±4 m) with the platforms found by Platform Levels and Falls, labelled with level and fall.

Grading quantities

Generated 2026-11-07 10:04 by LandXML Road & Terrain GIS Tools

To save as PDF, print this page from a browser (Chrome or Edge: Save as PDF). Pages are numbered and charts keep their colours; untick *Headers and footers* to hide the browser's file path.

Source: demo_grading.xml · volume surface: Demo Earthworks

Exact prism volumes of the volume surface's triangles, split at zero depth. Positive depth is fill and negative is cut. Volumes are in place, with no shrinkage or bulking factors.

Cut (m^3)	Fill (m^3)	Net, fill − cut (m^3)	Plan area (m^2)
7,210.74	8,239.54	+1,028.79	9,636.00

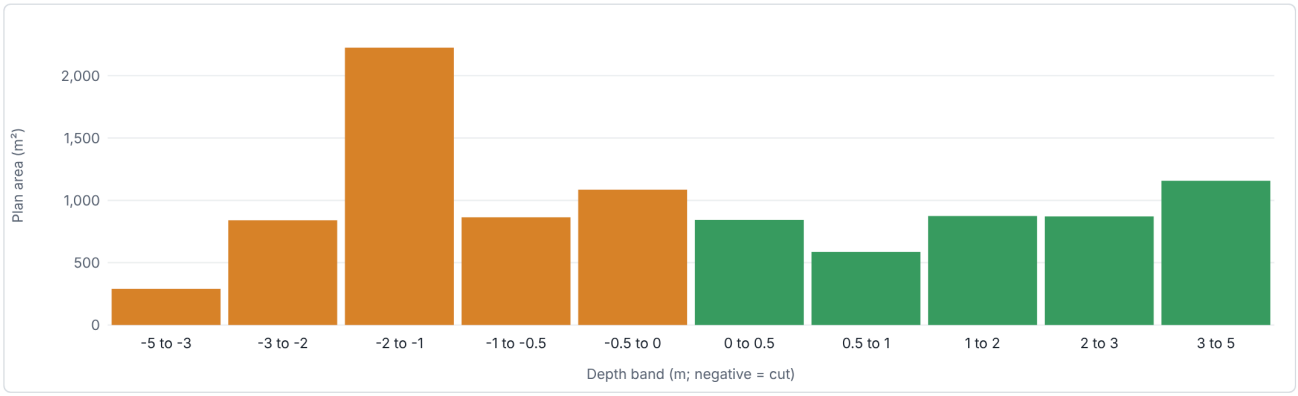
Quantities

Quantity	Value
Cut volume (m^3)	7,210.742
Fill volume (m^3)	8,239.536
Net, fill − cut (m^3)	1,028.794
Plan area (m^2)	9,636.000
Cut area (m^2)	5,305.103
Fill area (m^2)	4,330.897
At grade (m^2)	0.000
Maximum cut depth (m)	3.916
Maximum fill depth (m)	4.221
Triangles	4,818

Figure 8. Grading Quantities (Volume Surface) report: headline cut, fill, net and plan area, then the quantities table.

Plan area by depth (m)

Exact areas of the linear triangles between the depth limits; a band includes its lower limit.



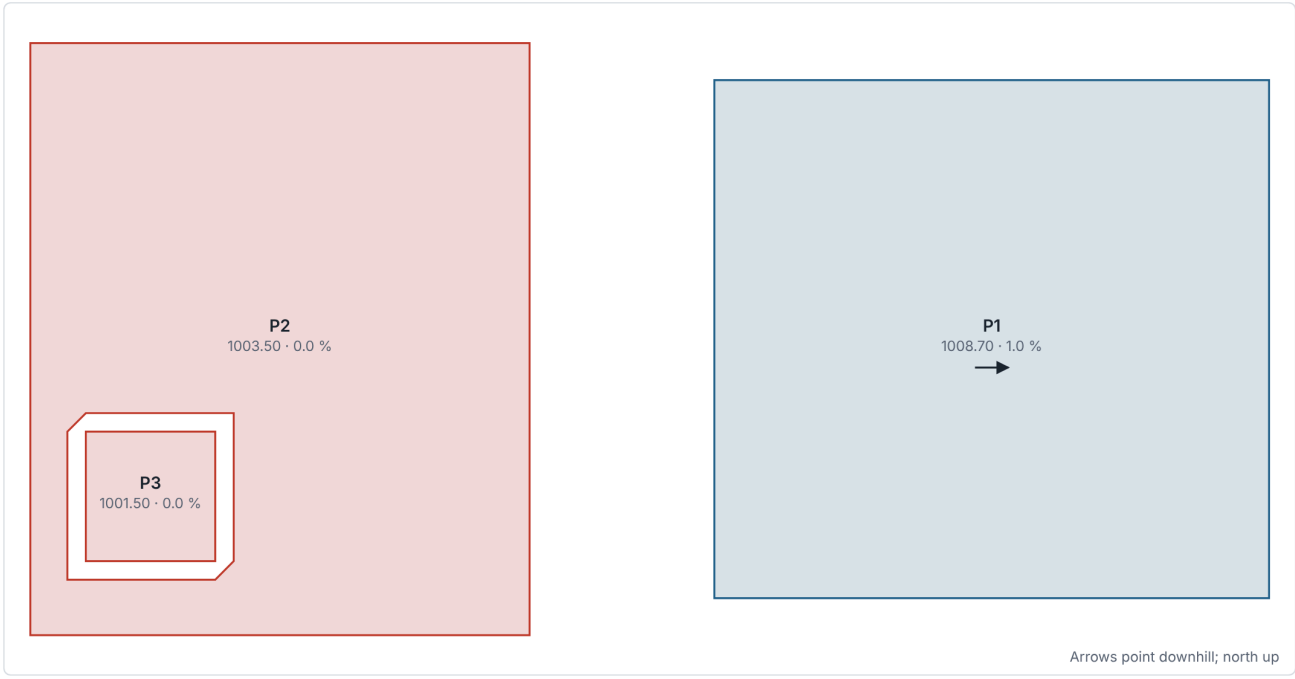
Depth band (m)	Area (m²)	Share (%)
< -5	0.00	0.00
-5 to -3	289.66	3.01
-3 to -2	840.34	8.72

Figure 9. Exact plan area in each depth band, cut bands in orange and fill bands in green.

Platform Levels and Falls

Run **Platform Levels and Falls** on the design (finished-level) surface. Each triangle is classed by slope: *flat* below the flat threshold (default 2 %), *sloping* up to 30 % (ramps and graded areas), *batter* from 30 % to 250 % (1V:3.3H to 1V:0.4H) and *near-vertical* above that (retained edges). Flat triangles that share an edge form a platform; platforms smaller than the listing limit (default 50 m²) are left out. Each platform is written as a polygon with its area, mean, minimum and maximum level, overall fall and the bearing it falls towards, mean local grade and the share of its area flatter than the minimum fall. The overall fall comes from an area-weighted plane fitted to the platform, so a platform built as two level terraces joined by a gentle transition reports its average fall; check the minimum and maximum levels. Platforms whose overall fall is below the minimum (default 0.5 %) are flagged, because water can pond on them; common targets are at least 1 % on gravel and 0.5 % on paved areas. The HTML report also gives the area in each slope class.

Platforms (3; 2 below minimum fall)



Meets minimum fall Below minimum fall

Platform	Area (m²)	Mean level (m)	Min (m)	Max (m)	Overall fall (%)	Falls towards	Mean local grade (%)	Area flatter than minimum (%)	Flag
P1	3,360.00	1,008.70	1,008.40	1,009.00	1.00	90°	1.00	0.00	
P2	3,136.00	1,003.50	1,003.50	1,003.50	0.00	–	0.00	100.00	below minimum fall
P3	196.00	1,001.50	1,001.50	1,001.50	0.00	–	0.00	100.00	below

Figure 10. Platform Levels and Falls: plan of the platforms with level, overall fall and a downhill arrow; level platforms are flagged in red. Holes such as the pond inside P2 are kept.

Earthwork Balance (volume surfaces)

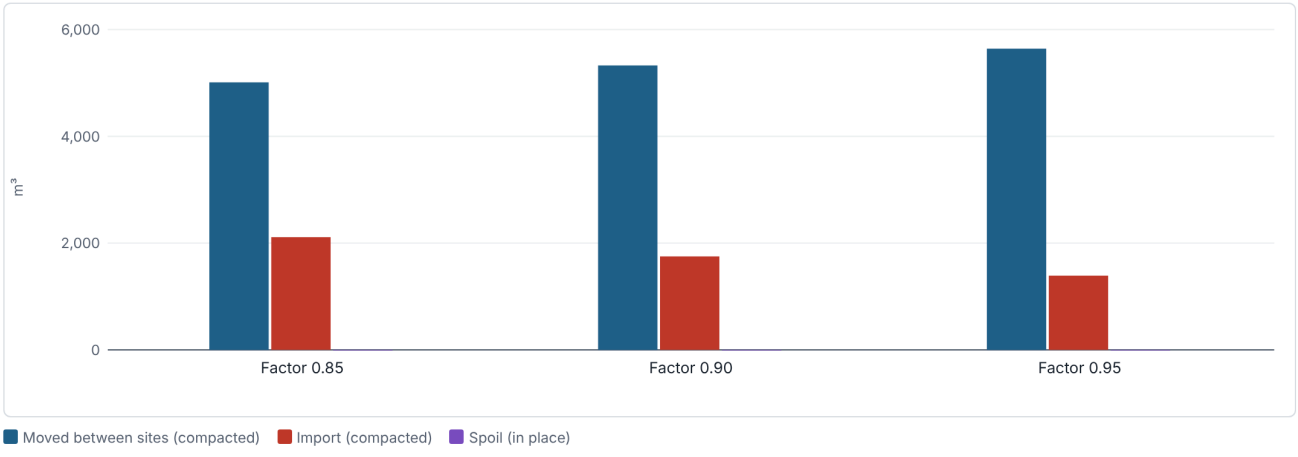
Run **Earthwork Balance (Volume Surfaces)** with the volume surfaces of the sites to balance, separated by semicolons, or leave the list blank to use every surface named as a volume surface. Cut and fill per site are the exact in-place volumes of Grading Quantities. For each shrinkage factor (compacted volume from one unit of in-place cut; default 0.85, 0.90 and 0.95) each site first uses its own cut; surplus from the other sites then supplies the shortfalls, assuming it can be hauled between them. The report gives, per factor, the volume moved between sites, the import still needed (compacted, and its in-place borrow equivalent at the same factor) and the spoil in place, with a table per site. An unsuitable share of cut is removed before balancing and goes to spoil; topsoil stripping (plan area × depth) is reported separately and not used as fill.

Sites (in place)

Site	Plan area (m²)	Cut (m³)	Fill (m³)	Net, fill − cut (m³)
Platform A Earthworks	4,796.00	6,319.96	361.84	-5,958.13
Platform B Earthworks	4,840.00	890.78	7,877.70	6,986.92
Total	9,636.00	7,210.74	8,239.54	1,028.79



Balance by shrinkage factor



Shrinkage factor	Moved between sites (m³, compacted)	Import (m³, compacted)	Import (m³, in place)	Spoil (m³, in place)

Figure 11. Earthwork Balance: cut and fill per site, then the volume moved between sites, import and spoil for each shrinkage factor.

Grading Revision Comparison

When a grading revision arrives, run **Grading Revision Comparison (Volume Surfaces)** with the volume surface of each revision; the second can come from another LandXML file (use the same coordinate interpretation and units). The table lists cut and fill of both revisions and the change (second minus first), overall and, with a zone polygon layer and its name field, by zone. Triangles are assigned to zones by their centroid, so zone totals near zone edges are approximate; the overall totals are exact. Labels such as "Rev B" and "Rev C" are used in the report. The optional change raster maps the depth change (positive = more fill or less cut); where only one revision has earthworks, the other counts as zero depth, so changed footprints show clearly. A change can also come from a revised existing-ground survey: compare the existing-ground surfaces with Surface Cut/Fill before attributing it to the design.

Overall

Zone	Rev A cut (m³)	Rev A fill (m³)	Rev B cut (m³)	Rev B fill (m³)	Cut change (m³)	Fill change (m³)	Net change, fill − cut (m³)
All	6,786.70	10,085.62	7,210.74	8,239.54	424.04	-1,846.08	-2,270.12

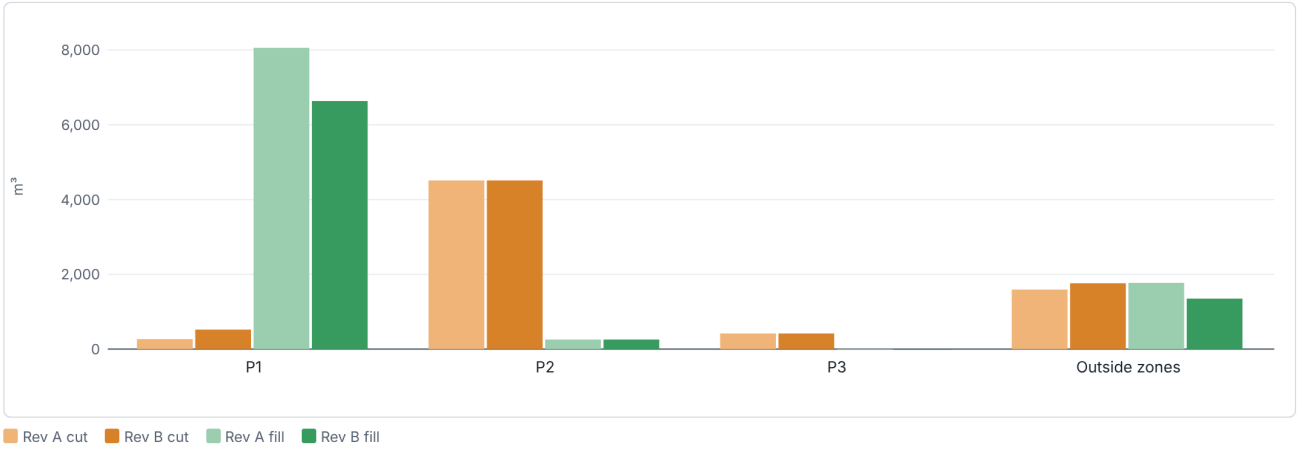


Figure 12. Grading Revision Comparison: overall change and cut and fill of both revisions by zone (here the platform polygons as zones).

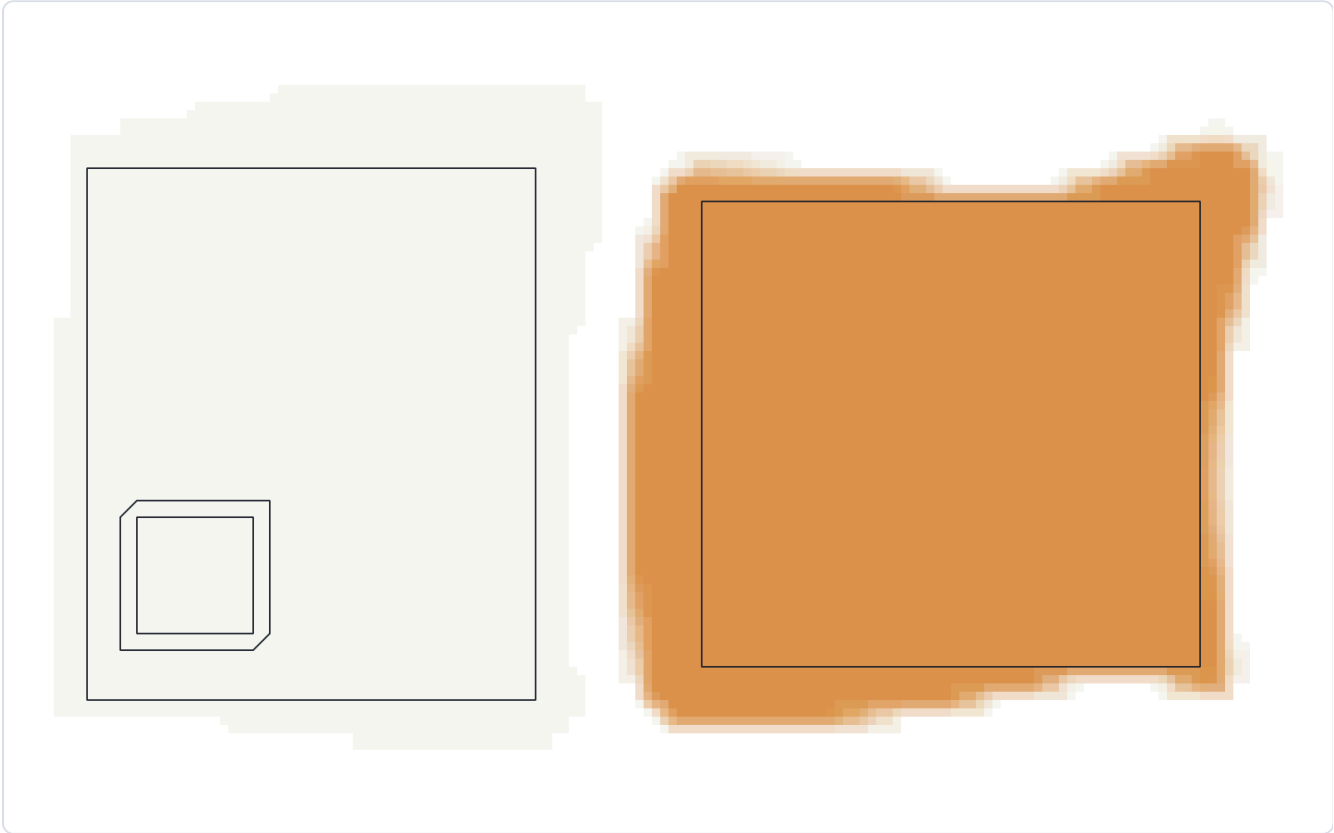


Figure 13. Depth-change raster (second revision minus first): platform B was lowered by 0.5 m, so it shows less fill (orange) while platform A is unchanged.

Basin Stage–Storage

Export the pond or basin grading as its own TIN surface and run **Basin Stage–Storage** on it. The storage below each level is integrated exactly over the triangles, because water depth is linear on each one; the table lists level, depth, water area and storage from the floor to the rim in the chosen step (default 0.25), with the freeboard level and the rim always included. Leave the rim blank to use the **spill level**, the lowest point on the surface's outer edge, where water would overflow; a rim above the spill level is flagged, because storage

above it exists only if the basin is closed beyond this surface. Enter the required volume in the reported units to check capacity with freeboard (rim minus freeboard, default 0.5): **PASS** when it fits with freeboard, **PASS TO RIM ONLY** when it fits only if filled to the rim, otherwise **FAIL**. Outputs are an HTML report and a CSV table.



Quantity	Value
Floor level (m)	1,001.500
Spill level, lowest point on the surface edge (m)	1,003.500
Rim level used (m)	1,003.500
Freeboard (m)	0.500
Storage to the rim (m³)	512.000
Water area at the rim (m²)	320.000
Storage with freeboard, to 1003.000 (m³)	360.375

Figure 14. Basin Stage–Storage: storage against level with the freeboard level, rim and required storage. Here 400 m³ fits only when the pond is filled to the rim (PASS TO RIM ONLY).

16 Cross-Section Sheets and Report

Plots corridor sections as multi-page PDF sheets (A4, A3 or A1; 1–12 sections per sheet) with a title block and legend, an HTML report with a section table and drawings, and one PNG per section. A station range and interval select what to plot, for example every 50 m between 10000 and 12000.

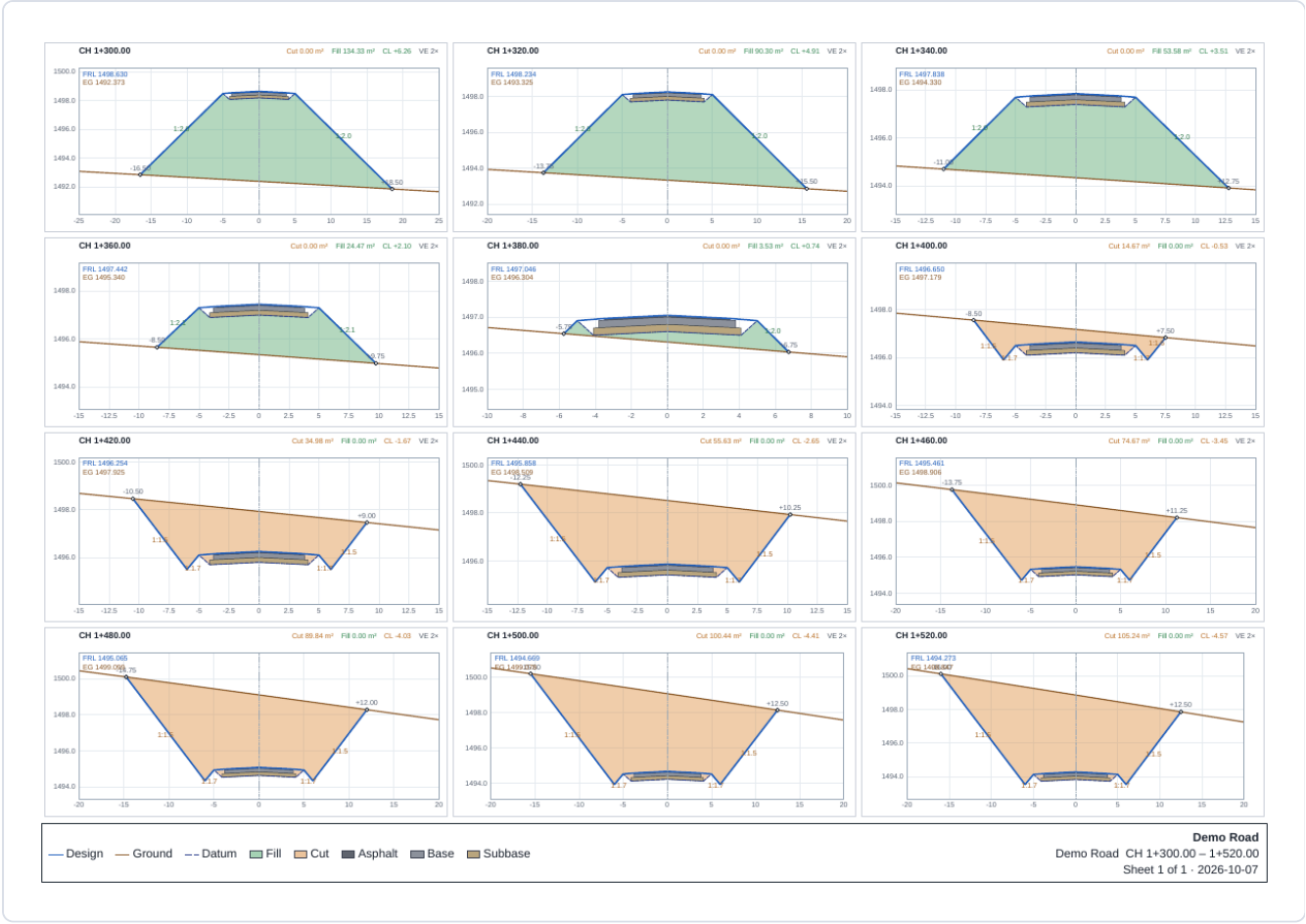
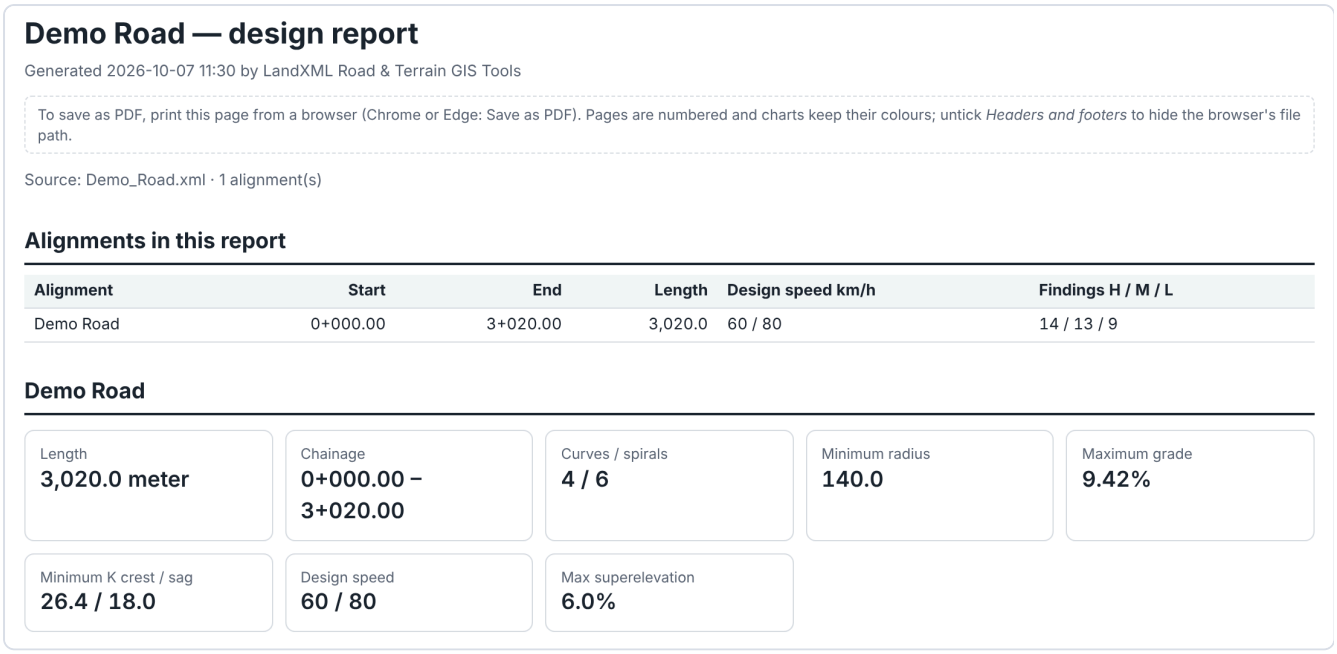


Figure 15. An A3 sheet with 12 sections across a fill-to-cut transition: ground and design, shaded cut and fill, pavement layers, Datum, slope ratios (red when steeper than the stabilisation limit), daylight offsets, FRL/EG levels and areas.

17 Road Design Report

Road Design Report writes one HTML page per file with a section per alignment. Print it to PDF from a browser for a paginated report; each alignment starts on a new page.



Demo Road — design report

Generated 2026-10-07 11:30 by LandXML Road & Terrain GIS Tools

To save as PDF, print this page from a browser (Chrome or Edge: Save as PDF). Pages are numbered and charts keep their colours; untick *Headers and footers* to hide the browser's file path.

Source: Demo_Road.xml · 1 alignment(s)

Alignments in this report

Alignment	Start	End	Length	Design speed km/h	Findings H / M / L
Demo Road	0+000.00	3+020.00	3,020.0	60 / 80	14 / 13 / 9

Demo Road

Length 3,020.0 meter	Chainage 0+000.00 – 3+020.00	Curves / spirals 4 / 6	Minimum radius 140.0	Maximum grade 9.42%
Minimum K crest / sag 26.4 / 18.0	Design speed 60 / 80	Max superelevation 6.0%		

Figure 16. The report opens with a table of its alignments (chainage, length, design speeds and review findings), followed for each alignment by summary cards: length, chainage, curves and spirals, minimum radius, maximum grade, minimum crest and sag K, design speed and maximum superelevation.

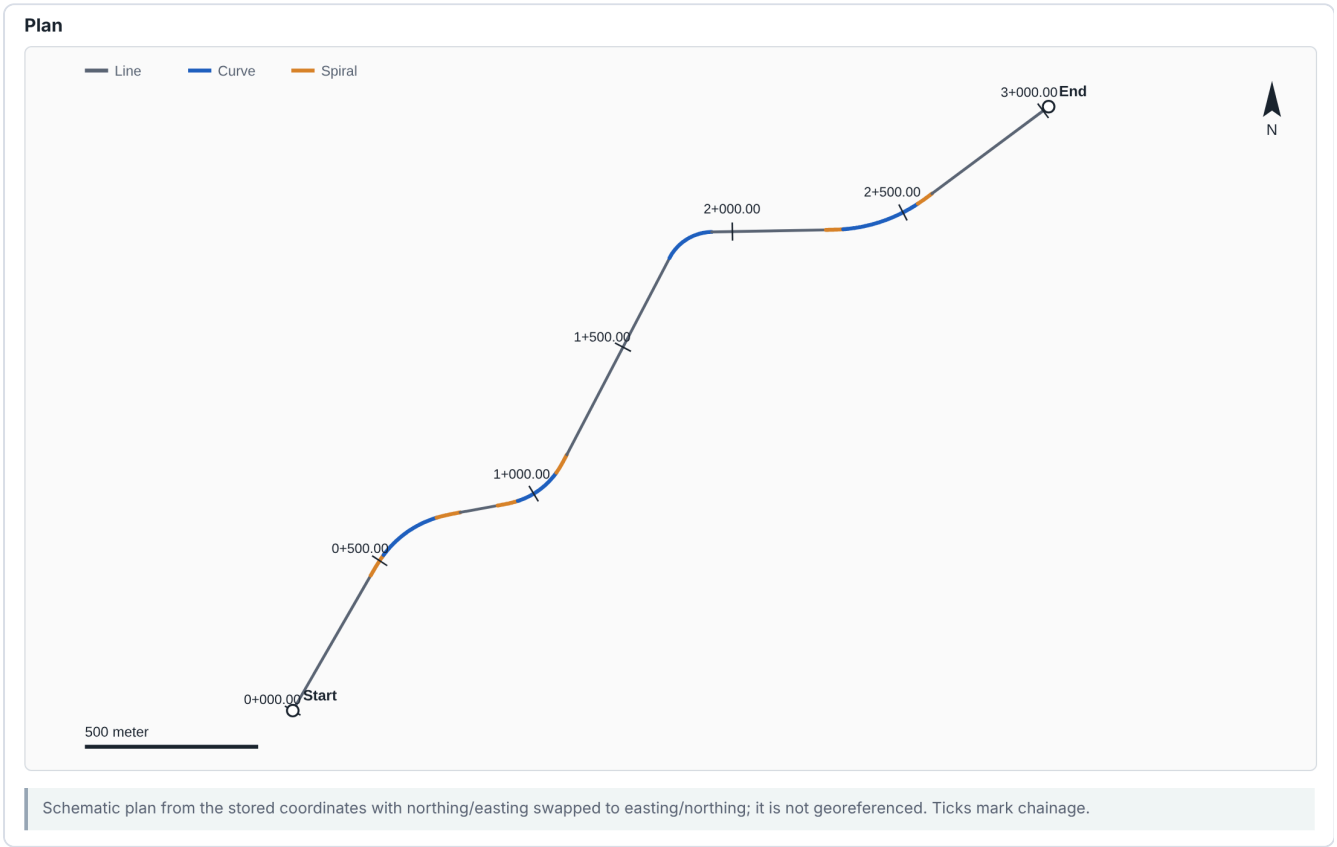


Figure 17. Schematic plan: lines, curves and spirals in different colours with chainage ticks, north arrow and scale bar. Choose whether the file stores northing first.



Figure 18. Curvature diagram (1000/R, right positive) with radius labels and spiral transitions.

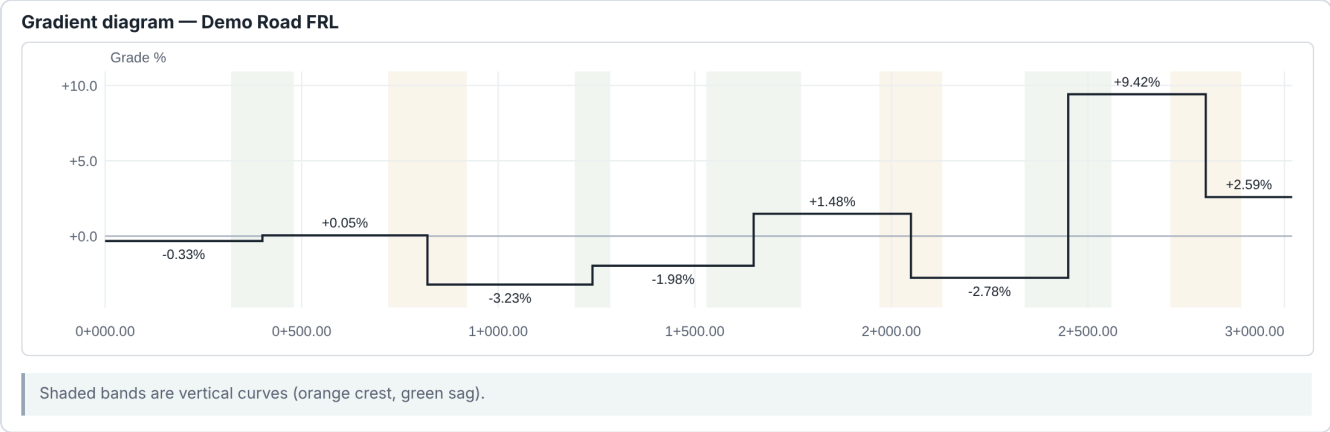


Figure 19. Gradient diagram with grade labels and crest/sag curve bands.



Figure 20. Superelevation diagram of both lanes from the transition stations and rates in the file. The normal crown is an input because LandXML does not store it.

The report also tabulates horizontal elements, vertical controls and superelevation transitions. Long alignments are split into strips (default 5,000 m per row). Select a **Design standard** to add the geometric design review (Part D) to each alignment.

PART D

Geometric design review

Check the horizontal and vertical alignment, superelevation and roadside against a design standard, and get computed changes to apply in the design software.

18 What the review does

Geometric Design Review reads the alignment, design profile, existing ground, superelevation, design speeds and corridor sections from the LandXML, applies the tables of the selected standard, and reports each problem with its location, the value found, the value required, the table it comes from and a recommendation. Where a fix is a profile change, it computes new PVI stations, elevations and curve lengths.

The review is limited to geometric design under the designer's control: alignment, profile, superelevation, side slopes and the design-speed plan. Pavement, drainage, structures, intersections and traffic operations are out of scope.

> **A DESIGN AID**
> The review flags and quantifies; it does not approve a design. Treat each recommendation as a starting point for the designer and record any Departure from Standards in the usual way.

19 Running a review

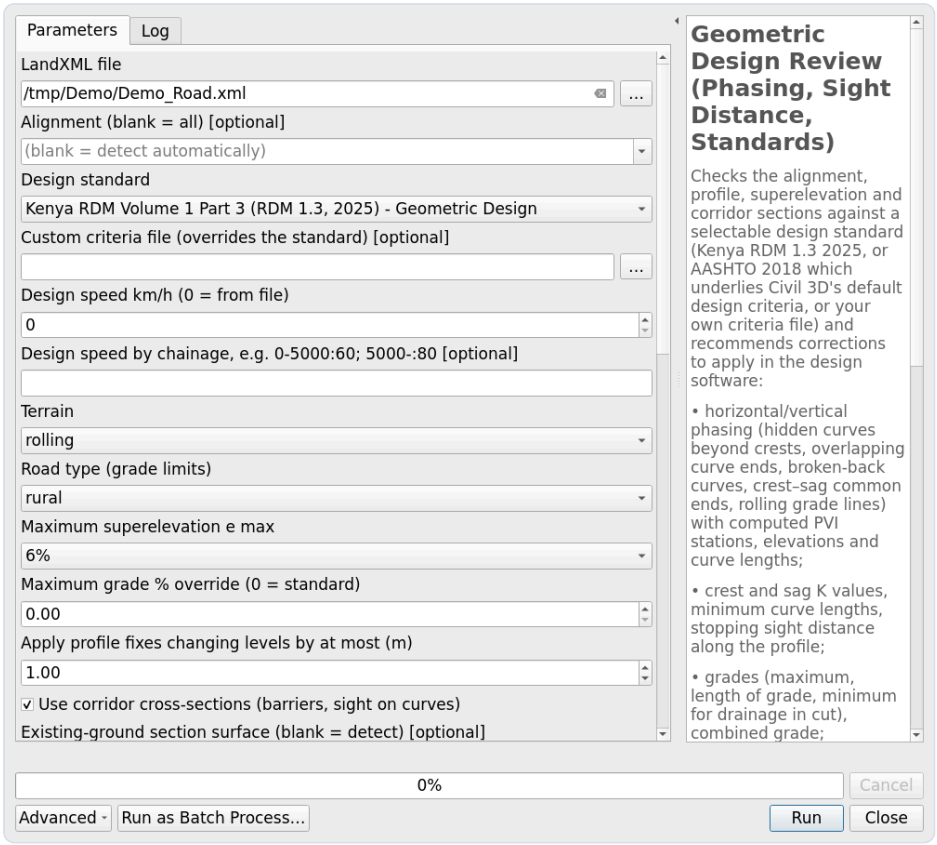


Figure 21. The Geometric Design Review dialog with the help panel.

Parameter	Meaning
Design standard	Kenya RDM 1.3 (2025) (default) or AASHTO 2018. See section 25.
Custom criteria file	A JSON criteria file that overrides the standard.
Design speed (0 = from file)	0 follows the speeds recorded in the file (recommended). A value applies one speed to the whole road.
Design speed by chainage	Overrides the file, e.g. `0-5000:60; 5000-:80`.
Terrain, Road type	Select the maximum-grade row (flat, rolling, mountainous, escarpment; rural or urban).
Maximum superelevation e max	4, 6 or 8 %, plus 10 % for AASHTO. The 10 % option uses Table 3-11: 15–80 mph for US customary and 20–130 km/h for metric. Other standards reject unsupported selections.
Maximum grade override	Required for AASHTO grade checks (AASHTO limits depend on road class).
Apply profile fixes ... at most (m)	Fixes changing any level by more than this are reported but not applied to the recommended profile (default 1 m).
Use corridor cross-sections	Enables barrier warrants and sight distance on curves.
Section surfaces	Ground and design section surfaces (blank = detect).

Parameter	Meaning
Outputs	Findings table, findings as map points (created by default; needs an output CRS and the same coordinate interpretation as your other layers), HTML report with a banded long section, plan and profile sheets (A3 PDF), recommended profile as Civil 3D text (to apply) and as a comparison LandXML, and a changed-PVI CSV.

Geometric design review — Kenya RDM Volume 1 Part 3 (RDM 1.3, 2025) - Geometric Design

Terrain rolling · rural · e max 6% · design speed from file

> Recommendations are computed values to apply and verify in the design software; the source design is not changed. Each finding cites its criteria table. Only geometric elements under the designer's control are checked.

High (safety / absolute minimum)	Medium (desirable / standard)	Low (aesthetic / advisory)	Profile changes proposed
14	13	9	3

Category	High	Medium	Low
Design speed	0	1	1
Grades	1	1	1
Horizontal	2	5	4
Phasing	2	1	3
Roadside safety	6	3	0
Superelevation	2	1	0
Vertical	1	1	0

Figure 22. Review summary: findings by severity and category, and the number of profile changes applied to the recommended profile.

Severity	Meaning
High	Below an absolute minimum or a safety requirement: stopping sight distance, minimum K or radius, barrier warrant, abrupt speed change, hidden curve.
Medium	Below a desirable or standard value, or needing an engineering assessment.
Low	Aesthetic or advisory: appearance of the alignment, preferred values, short sections.

20 Design speed and speed changes

Most checks depend on design speed, so the review follows the speeds recorded along each alignment: Civil 3D `SpeedStation` features (set in **Alignment Properties → Design Criteria** before export) or Roadway `DesignSpeed` records. Where two speeds share a station, the later entry applies, as in Civil 3D.

- Each element (curve, vertical curve, straight) is checked at the **highest** speed over its length, so an element spanning a speed change meets the faster approach.
- The report tabulates the speed sections with their lengths and changes. When the whole road has one speed, it reminds you to define speed stations where the terrain changes.
- The log says when file speeds are ignored because you entered a speed or ranges.

Design speed sections

From	To	Length	Design speed km/h	Change km/h
0+000.00	1+900.00	1,900	80	—
1+900.00	3+020.00	1,120	60	-20

Figure 23. Design speed sections of the demo road: 80 km/h then 60 km/h from CH 1+900.

The speed plan itself is checked (Kenya RDM 1.3 §3.10.3, §5.5, §5.14):

Code	Check and recommendation
D1	Speed step above 20 km/h (high), or 10–20 km/h (low; acceptable but not preferred). The recommendation lists intermediate sections in 10 km/h steps with chainages, placed on the faster side so the slower section's design remains valid, e.g. "90 km/h over 0–1,000; 80 km/h over 1,000–2,000; 70 km/h over 2,000–3,000".
D2	A transition section shorter than 1 km, or a short section between two sections of the same speed (speed oscillation). A short faster section should be merged; elements in a short slower section should be designed for the approach speed.
D3	A curve sharper than the faster speed allows within 1 km after a speed reduction (in either direction of travel). Move the speed change, add an intermediate section or increase the radius.

21 Checks and finding codes

Code	Check	Kenya RDM 1.3 reference
V0	Grade change without a vertical curve	Tables 6.1, 6.3, 6.4
V1	Crest or sag K below minimum (with the length needed)	Table 6.1 (crest), Table 6.4 (sag)
V2	Vertical curve shorter than the minimum length, including small grade changes (A < 0.5 %: L ≥ 2V)	Table 6.3, S7.3
G1–G2	Grade above the absolute or desirable maximum for terrain and road type	Table 6.5
G3	Grade longer than its critical length	Table 6.6
G4	Grade flatter than 0.5 % in cut (drainage)	S6.5.4
S1	Stopping sight distance along the profile	Tables 3.13, 3.16
S2	Sight distance on horizontal curves blocked by cut slopes	Table 3.16
H0–H1	Radius below the absolute floor or the minimum for speed and e max	S5.8, Table 5.1
H2–H3	Missing transition spiral, or spiral parameter A too small	Table 5.8, Eq. 5.4
H4–H5	Curve too short, or too long	Table 5.5
H6	Broken-back curves (short straight between same-direction curves)	S5.6, S7.8
H7–H8	Short straight between reverse curves; long straight	Table 5.4
H9	Isolated sharp curve after a long straight (needs 1.5 × minimum radius)	S5.4
E1–E3	Superelevation missing, below the required rate, or above e max	Tables 5.9–5.11

Code	Check	Kenya RDM 1.3 reference
E4	Superelevation run-off too short	Table 5.7, S5.9
E5	Combined grade √(g² + e²) too steep	S6.5.3
P1–P7	Alignment phasing (section 22)	Chapter 7
R1	Embankment barrier warrant (section 23)	S12.6.5, Figure 12.4, Table 12.6
D1–D3	Design speed changes (section 20)	S3.10.3, S5.5, S5.14

Findings and recommendations

Severity	Chainage	Issue	Recommendation	Reference
Design speed · 1 medium, 1 low				
25. Medium D3	1+790.00 1+940.00	Sharp curve R140 close to a speed reduction (80 → 60 km/h) The curve spans the speed change, so drivers enter it at the higher speed; R140 is below the 80 km/h minimum (265 m).	Move the speed change at least 1000 m before the curve, add an intermediate speed section, or increase the radius to ≥ 265 m.	S3.10.3, S5.5, S5.14
34. Low D1	1+900.00	Design speed change 80 → 60 km/h A change of 20 km/h is acceptable but steps of 10 km/h or less are preferred.	Consider an intermediate 70 km/h section of at least 1000 m.	S3.10.3, S5.5, S5.14
Grades · 1 high, 1 medium, 1 low				
14. High G1	2+560.00 2+710.00	Grade +9.42% exceeds the absolute maximum Absolute maximum 6% (rural, rolling).	Flatten the grade, lengthen the development or record a Departure from Standards.	Table 6.5
15. Medium G4	0+000.00 0+320.00	Flat grade -0.33% in cut Below the 0.5% minimum for longitudinal drainage over 320 m.	Use at least 0.5% or provide side drains with independent falls.	S6.5.4
28. Low G4	0+480.00 0+720.00	Flat grade +0.05% Below the 0.5% minimum for longitudinal drainage over	Use at least 0.5% or provide side drains with independent falls.	S6.5.4

Figure 24. Findings table (first rows), grouped by category: severity and code, chainage range, issue and detail, recommendation and the criteria reference.

22 Alignment phasing and recommended PVIs

Phasing compares horizontal and vertical curves so the road reads clearly to drivers and looks right. Following RDM 1.3 Chapter 7, the review reports:

Code	Situation
P1	The start of a horizontal curve is hidden beyond a crest, within stopping sight distance of the crest: the driver cannot see the change of direction in time.
P2	A crest or sag overlaps both ends of a horizontal curve without matching it.
P3	A vertical curve overlaps only one end of a horizontal curve.
P4	A short crest inside a sharp horizontal curve.
P5	Horizontal and vertical curves separated by less than max(SSD/2, 30 m): separate them further or make them concurrent.
P6	Rolling grade line: a crest and a sag within one horizontal curve; keep a constant grade through the curve.
P7	Crest and sag without the 60–100 m of constant grade between them (rolling grade line).

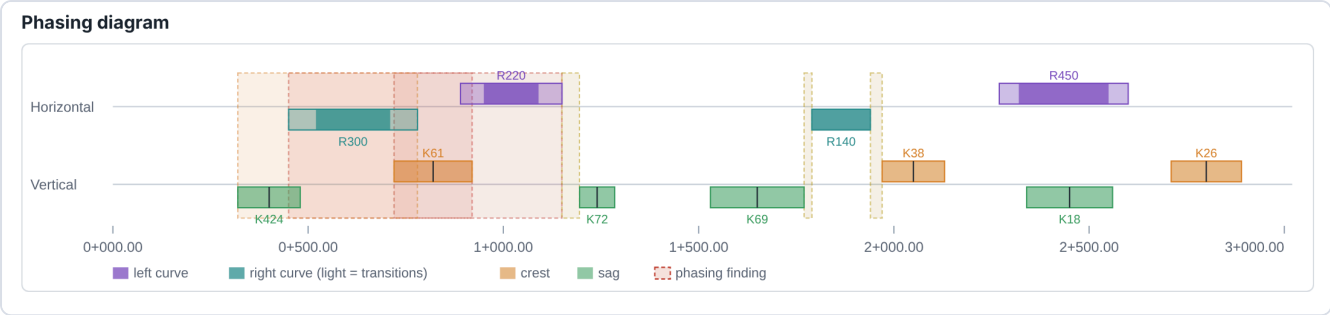


Figure 25. Phasing diagram: horizontal curves (left purple, right teal, light ends = spirals) above vertical curves (crest orange, sag green) on a common chainage; dashed boxes mark phasing findings.

How fixes are computed

For each phasing, K or curve-length finding the tool tries these profile changes and keeps the feasible one that changes levels least:

1. **One end coincident**: change the vertical curve length so one end meets the horizontal curve's start or end (only when the horizontal curve is not sharp, R ≥ 1.5 R_{min}).
2. **Separation**: shorten the vertical curve to clear the horizontal curve by at least max(SSD/2, 30 m) (not used for hidden curves, P1).
3. **Both ends coincident**: move the PVI to the middle of the horizontal curve and match its length; the preferred arrangement in Chapter 7.

Each candidate must meet minimum K, avoid overlapping neighbouring curves and stay within grade limits. Fixes that change any level by more than the cap are reported with the change they need (redesign that section) and are not applied.

Recommended PVIs (● changed)

PVI	Elevation	Curve length	Grade in %	Grade out %
0+000.00	1,514.500	–	–	-0.199
0+615.00 ●	1,513.275	210.000	-0.199	0.046
0+820.00	1,513.369	200.000	0.046	-3.226
1+240.00 ●	1,499.818	140.000	-3.226	-1.980
1+650.00	1,491.699	240.000	-1.980	1.477
2+050.00	1,497.606	160.000	1.477	-2.783
2+450.00 ●	1,486.474	235.000	-2.783	9.424
2+800.00	1,519.458	180.000	9.424	2.595
3+020.00	1,525.166	–	2.595	–

Figure 26. Recommended PVIs with changed rows marked ●: station, elevation, curve length and grades.

Long section and plan & profile sheets

The report draws a banded long section for each window of **Sheet and long-section window length** (default 1,000 m): the design profile, the recommended profile dashed where fixes apply, the existing ground, changed PVIs, and every finding as a numbered marker over its shaded chainage range. Below the profile, the bands show the design speed, the horizontal curves (left above the line, right below), the vertical curves (crest above, sag below) and the tangent grades. Horizontal and vertical curves sit in adjacent bands on one chainage, so their phasing reads directly; phasing findings (P1–P7) are outlined across both bands. Finding numbers are the same in the findings table, the `finding_no` field of the findings layer, the report and the sheets.

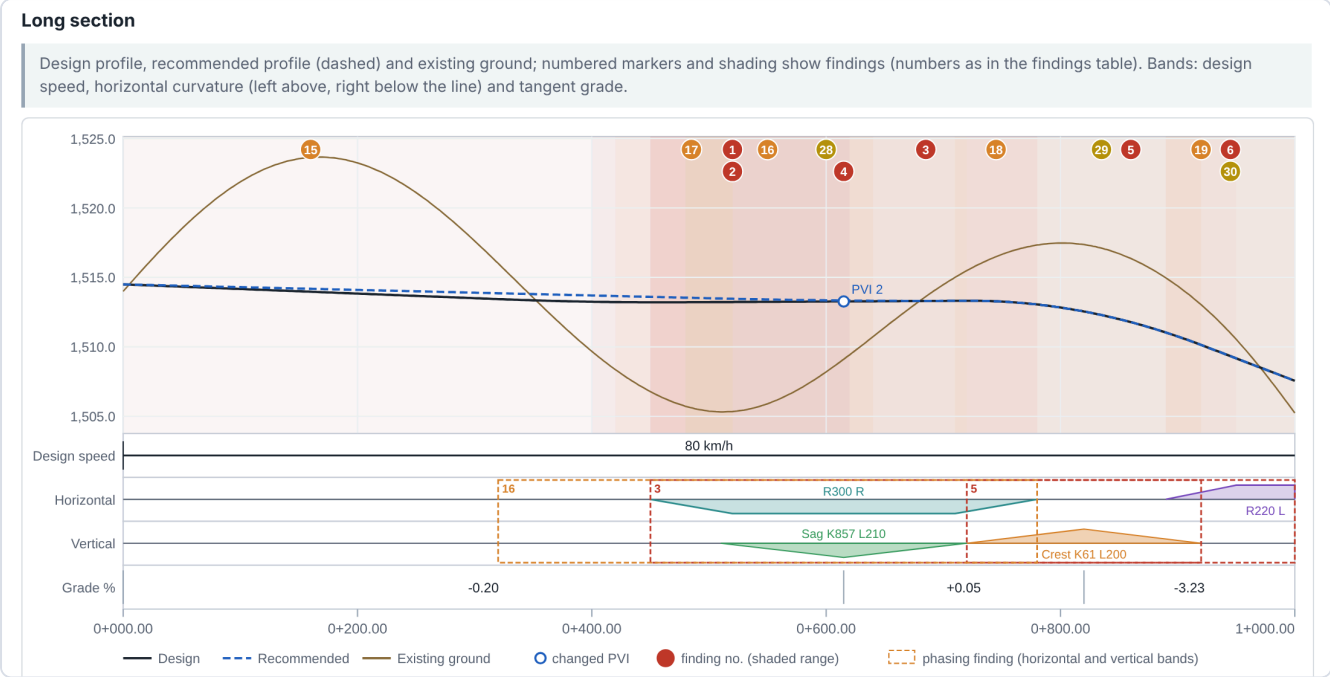


Figure 27. Banded long section in the review report: profiles and numbered findings, then design speed, horizontal and vertical curves (phasing findings dashed) and grades.

Plan and profile sheets (A3 PDF) puts one window on each A3 landscape sheet: a plan strip of the alignment with chainage ticks and the numbered findings beside it, the same banded long section with a findings lane that stacks overlapping ranges, and a table of the findings in that window (highest severity first; any beyond the table are counted and listed in the report). The plan is schematic and rotated so each window runs left to right; the north arrow shows grid north of the source coordinates. With an output CRS the chosen coordinate interpretation decides the X/Y order; without one, northing-first files are detected for the drawing only. Each sheet has a single border, a floating legend, notes, and a title box at the bottom right with the alignment, chainage range and sheet number.

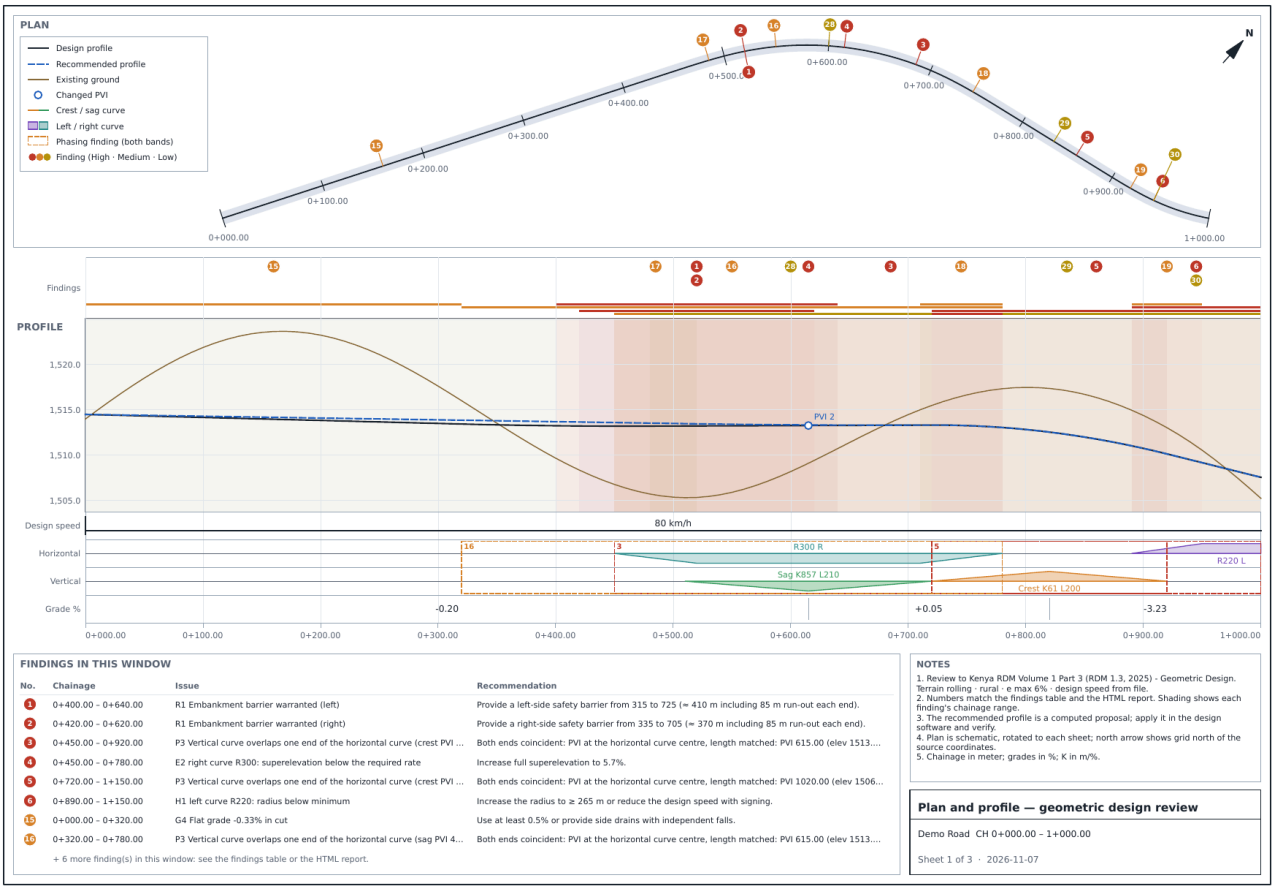


Figure 28. A plan and profile sheet of the demonstration road (first 1,000 m).

23 Safety barriers and sight on curves

Barriers (R1). From each corridor section the tool measures the embankment height from the shoulder edge to the toe and the side-slope ratio on each side. Under RDM 1.3 a barrier is warranted where the fill is at least 6 m high and steeper than 1:3 (the example stated with Figure 12.4); fills of 3–6 m steeper than 1:3, and fills of 6 m or more with slopes between 1:3 and 1:4, are reported for assessment against the chart. Adjacent sections are grouped into runs, and the recommendation gives the barrier extent per side with the run-out length of Table 12.6 added at each end.

Safety barrier schedule

Side	From	To	Status	Max fill height	Recommendation
left	0+400.00	0+640.00	warranted	9.57	Provide a left-side safety barrier from 315 to 725 (≈ 410 m including 85 m run-out each end).
right	0+420.00	0+620.00	warranted	6.53	Provide a right-side safety barrier from 335 to 705 (≈ 370 m including 85 m run-out each end).
left	1+020.00	1+320.00	warranted	10.55	Provide a left-side safety barrier from 935 to 1405 (≈ 470 m including 85 m run-out each end).
right	1+020.00	1+340.00	warranted	12.28	Provide a right-side safety barrier from 935 to 1425 (≈ 490 m including 85 m run-out each end).
right	2+520.00	2+740.00	warranted	6.23	Provide a right-side safety barrier from 2465 to 2795 (≈ 330 m including 55 m run-out each end).
left	2+540.00	2+740.00	warranted	6.16	Provide a left-side safety barrier from 2485 to 2795 (≈ 310 m including 55 m run-out each end).
right	1+720.00	2+100.00	to be assessed	5.92	Assess against the barrier warrant chart; Provide a right-side safety barrier from 1635 to 2185 (≈ 550 m including 85 m run-out each end).
left	1+800.00	1+860.00	to be assessed	3.19	Assess against the barrier warrant chart; Provide a left-side safety barrier from 1715 to 1945 (≈ 230 m including 85 m run-out each end).
left	1+980.00	2+000.00	to be assessed	3.03	Assess against the barrier warrant chart; Provide a left-side safety barrier from 1925 to 2055 (≈ 130 m including 55 m run-out each end).

Figure 29. Safety barrier schedule by side, with the extent including run-out.

Sight on curves (S2). On horizontal curves the review computes the sight-line offset needed for stopping sight distance and checks whether existing ground beyond the cut daylight line rises above the sight line (at the height set by the standard) on the inside of the curve. Only ground beyond daylight counts, so excavated material does not create false obstructions.

> **FIGURE 12.4**
> The barrier warrant chart in RDM 1.3 is an image; only its stated example is encoded. Treat findings near the thresholds as prompts for engineering judgement.

24 Taking recommendations back to Civil 3D

The recommended profile is written in three forms, each for a different job. The profile files are complete profiles: the original PVIs with the applied changes, so the PVI count does not change.

To	Use
Apply the fixes to the design alignment	The text file, with *Create Profile from File* on that alignment (below).
Compare the design and recommended profiles	The comparison LandXML. Civil 3D imports it as a **new** alignment, named `… (review)`, carrying both profiles.
See which PVIs changed	The changed-PVI CSV: PVI number, old and new station, level and curve length.

Text file for Create Profile from File

```
0.000 1514.500
615.000 1513.275 210.000
820.000 1513.369 200.000
1240.000 1499.818 140.000
…
3020.000 1525.166
```

1. In Civil 3D choose **Home → Profile → Create Profile from File**.
2. Select the alignment and the `.txt` file. Each line is station, elevation and, for intermediate PVIs, the parabolic curve length.
3. Compare the new profile with the original in the profile view; adjust and re-export to review again.

Comparison LandXML

The comparison LandXML holds a copy of the alignment geometry, renamed `… (review)`, with two profiles: the design profile under its own name and the recommended profile named `… recommended (standard)`. Civil 3D's **Insert → LandXML** import always creates new alignments, so importing this file into the design drawing gives a separate review alignment rather than changing the design one. Draw both of its profiles in one profile view to compare them, or load the file into the Profile Viewer. Do not superimpose the review alignment's profile onto the design alignment to apply the fixes: use the text file for that.

Changed-PVI CSV

One row per PVI whose station, level or curve length changed, numbered from 1 along the profile as in Civil 3D's PVI table, with old and new values. Check it against the recommended PVIs table in the report before

applying the text file.

> **ITERATE**
> Run the review, apply the changes you accept in Civil 3D, export again and re-run. Findings that need more than the level-change cap are a sign that the section needs redesign rather than adjustment.

25 Design standards and custom criteria

Two standards ship with the plugin as editable JSON files in its `criteria` folder:

- **Kenya RDM 1.3 (2025)** — Road Design Manual Volume 1 Part 3: Geometric Design of Highways, Rural Roads and Urban Roads. Values are taken from the manual's tables and every finding cites the table used. This is the default.
- **AASHTO 2018** — A Policy on Geometric Design of Highways and Streets (metric), the basis of Civil 3D's default design criteria. AASHTO maximum grades depend on road class and are not tabulated, so enter a **Maximum grade** for grade checks. Barrier run-out lengths are not included; the warrant still runs. Speed-change steps of 10 and 20 km/h are used as guidance without a minimum transition length.

To use another standard, copy one of the files, change `id` and `name`, edit the tables and select it as **Custom criteria file**. Tables keyed by design speed are interpolated linearly between tabulated speeds and held constant beyond the ends. Only metric criteria are accepted.

Key	Content
`ssd`, `ssd_by_grade`	Stopping sight distance by speed, optionally by grade.
`eye_height`, `object_height`, `sightline_height`	Heights for sight-distance checks.
`k_crest`, `k_sag`, `min_vc_length`	Vertical curve minimums.
`max_grade`, `max_grade_length`, `min_grade_percent`	Grade limits by road type and terrain; critical lengths.
`min_radius`, `side_friction`	Minimum radius by e max, or computed from side friction.
`superelevation_rates`	Required rate by radius and speed for each e max (NC, RC or a rate).
`runoff_min`, `transition_required_radius`, `spiral_a_coefficient`	Run-off and transition rules.
`tangent_min_factor`, `tangent_max_factor`, `same_direction_tangent_min`	Straight lengths.
`crest_sag_gap`, `small_grade_change`	Phasing and small-grade-change rules.
`barrier`	Warrant height and slope, assessment band, run-out by speed, minimum length.
`speed_change`	Preferred and maximum speed steps, minimum transition length.
`references`	Table numbers printed with each finding.

PART E

Technical reference

How results are computed, what each output contains, what LandXML content is supported, and how the plugin has been validated.

26 Computation methods

Stationing

Chainage is true arc length along lines, circular curves and clothoid spirals (curvature varying linearly with length). On a 36 km test alignment the end chainage matches the declared length to 0.1 mm; earlier chord-based stationing drifted by 0.33 m.

Profiles

Design profiles are evaluated from PVIs and symmetric parabolic curves: $z = z_{VPC} + g_1 x + (g_2 - g_1)x^2/2L$. $K = L / |A|$ with A in percent. Existing-ground profiles are linear between points; gaps in coverage are kept as gaps.

Earthworks

Section areas come from the polygon between the ground and design lines (to the Datum where used). Volumes use average end areas, $V = (A_1 + A_2)/2 \cdot d$, between sections no farther apart than the maximum spacing. The mass-haul ordinate is the running sum of factored cut minus fill. Side-slope areas are the sloped lengths from hinge to daylight, averaged between adjacent sections and multiplied by their spacing.

TIN rasterization and difference

Each cell centre takes the elevation of the visible triangle containing it, by barycentric interpolation. TIN difference grids both surfaces on the same cells and sums (compare − base) × cell area over cells where both exist.

Exact volumes from a volume surface

With depths sorted $a \le b \le c$ and plan area A, a triangle's signed volume is $A(a + b + c)/3$. When the depth changes sign, the sub-triangle at the lone vertex has plan fraction $(a/(a - b))(a/(a - c))$ and volume $A \cdot \text{fraction} \cdot a/3$ (using c for a lone positive vertex), which gives cut and fill separately. The plan fraction below a depth t is $(t - a)^2/((b - a)(c - a))$ for $a < t \le b$ and $1 - (c - t)^2/((c - a)(c - b))$ for $b < t < c$, which gives exact depth-band areas.

Stopping sight distance on the profile (S1)

The profile is sampled every 2 m. From each station, in both directions, the line from the driver's eye (eye height above the road) to an object (object height) is extended until the road surface rises above it; the distance reached is compared with the SSD required at that speed and grade.

Superelevation lookup

For each curve the required rate is read from the table for the chosen e max, using the next smaller tabulated radius and the next higher speed column (the conservative reading); *NC* and *RC* mean normal crown and reverse crown. Where a standard has no table, e = $V^2/127R - f$.

Barrier warrant

Fill height is measured from the shoulder edge to the toe of the embankment; the side-slope ratio is horizontal over vertical between them. Runs of warranted sections are merged and extended by the run-out length.

27 Output layers and fields

Design review table and map points

Field	Content
`alignment_name`	Alignment.
`severity`, `category`, `code`	High/Medium/Low; Vertical, Grades, Sight distance, Horizontal, Superelevation, Phasing, Roadside safety, Design speed; finding code.
`sta_start`, `sta_end`, `from_label`, `to_label`	Chainage range (numbers and formatted labels).
`side`	left/right for barriers and curve sight.
`issue`, `detail`, `recommendation`, `reference`	Text of the finding and the criteria table.
`value`, `required`	Value found and value required (units as stated in the detail).

Profile Cut/Fill station table

`alignment_name`, `design_profile`, `ground_profile`, `station`, `station_label`, `kind`, `design_elev`, `ground_elev`, `depth` (design − ground).

Corridor quantities table

`station`, `section_name`, `ground_surface`, `design_surface`, `cut_area`, `fill_area`, `centre_depth`, `left_daylight`, `right_daylight`, `left_hinge`, `right_hinge`, `fill_slope_len`, `cut_slope_len`, `steep_len`, `datum_used`, and one `m_<shape>` area field per corridor shape.

Side-slope polygons

`side`, `kind` (fill/cut/mixed), `sta_start`, `sta_end`, `fill_slope_area`, `cut_slope_area`, `steep_area`, `plan_area`.

28 Supported LandXML content and limits

Element	Support
`Surface/Definition` (TIN)	Points and faces; invisible faces excluded; several surfaces per file.
`SurfVolume`	Recorded Civil 3D volumes shown beside TIN difference results.

Element	Support
`Alignment/CoordGeom`	Line, Curve, clothoid Spiral. Other spiral types and disconnected elements are reported, not drawn.
`ProfAlign`	PVI and ParaCurve. Circular vertical curves are reported as unsupported.
`ProfSurf`	Existing-ground and other surface profiles, with gaps.
`CrossSects`	Absolute `PntList3D` and station-relative `PntList2D` sections; `DesignCrossSectSurf` shapes, links and Datum.
`Superelevation`	Transition stations and full rates (two-lane rotation about the centreline assumed).
`Feature name="SpeedStation"`, `Roadway/DesignSpeed`	Design speeds by station.
`Breaklines`, feature lines	Geometry with names, codes and IDs.

Not supported: decreasing station progression and specialized railway posting, pipe networks, COGO points and plan features, DWG/DGN files, curvature correction of volumes, intersection and passing sight distance. XML is parsed with a hardened parser that refuses DTDs and entity expansion.

29 Validation

- **Automated tests:** 249 tests (parser, earthworks, site grading, design review, criteria tables, Processing algorithms and offscreen GUI) pass on QGIS 3.44.15 and 4.2.3 with synthetic fixtures whose quantities have closed-form answers.
- **GitHub Actions:** pull requests and pushes to `main` run the full synthetic suite in headless QGIS 3.44.15 and 4.2.3 containers, including Processing, preparation, GUI, project settings, styles and station/offset locator checks. Desktop checks and verification against survey control remain necessary.
- **Civil 3D volumes:** on two private exports, TIN-difference volumes match the volumes Civil 3D recorded in the same files to within 0.1 m³ (for example fill 1,358.8 m³ recorded, 1,359.0 m³ computed).
- **Basin storage:** on two pond surfaces from the same exports, storage to the rim and with 0.5 m freeboard reproduced the values from an independent calculation, including one basin that meets its required volume only when filled to the rim.
- **Platform levels:** on the same two exports, platform polygons were valid and their areas equal to the summed triangle areas; level platforms and those with 1–1.8 % falls were identified as designed.
- **Grading quantities:** on two private industrial-site exports, Grading Quantities reproduced the Civil 3D cut/fill summaries within 0.005 m³ and the plan areas within 0.003 m², the rounding of the Civil 3D tables (0.6 and 0.9 million triangles).
- **Depth surface from two TINs:** on the same project, the depth surface built from the design (9,206 triangles) and the existing ground (1.2 million triangles) reproduced the Civil 3D volume surface and summary exactly to 0.001 m³, in about 6 s. For the second site, an existing ground rebuilt from the Civil 3D volume surface reproduced it within 0.001 m³; the separately exported existing ground there differed from the one used in that revision.
- **Independent methods:** on a 16.7 km corridor, TIN difference gave 67,282 m³ of cut against 67,056 m³ from the corridor sections.
- **Design review:** run on six private Civil 3D exports (9 short to 36 km alignments) under both standards without errors; findings were checked against the manual's tables, and the phasing fix was verified on

synthetic cases.

Project names are withheld; the validation notes are in `docs/earthworks-validation.md` in the repository. Validation of the plugin is not survey-control validation of a particular project: check placement and levels against control.

30 Quality-control checklist

☐ Inspect LandXML report saved; units and axis order understood.
☐ Coordinate interpretation and output CRS recorded and checked against survey control.
☐ Alignment lengths match the design; no geometry reported as skipped.
☐ Profiles cover the expected chainage; gaps in existing ground understood.
☐ DEM extent and elevation range plausible; hidden triangles excluded.
☐ Cross-section surfaces correctly identified (check the log).
☐ Quantities compared with the design software's reports before use.
☐ Review run with the correct standard, terrain, e max and design speeds.
☐ Each high finding resolved or recorded as a Departure from Standards.
☐ Source LandXML, parameters and outputs archived together.

31 Troubleshooting

Symptom	Check
Geometry in the wrong place or mirrored	Coordinate interpretation (Civil 3D usually needs Swap X / Y), offsets, CRS.
"Unit does not match the CRS"	Choose an output CRS in the unit the file declares (metres, feet or US survey feet).
Alignment element skipped	The element does not match its declared length or connect to its neighbour; check the export.
No cross-sections	The export has no sample-line sections, or only template data. Run Inspect LandXML.
Section surfaces not identified	Enter the ground and design section surface names from the dropdowns.
"The file records no design speed"	Define design speeds in Civil 3D before export, or enter a speed or ranges.
Very many findings	Check that the design speed and terrain match the road; use speed ranges where terrain changes. Read the category summary first.
Grade checks missing under AASHTO	Enter a maximum grade.
Cannot find the Profile Viewer	It is not in the Processing Toolbox: use **Plugins → LandXML Road & Terrain** or the plugin's toolbar button; the panel opens along the bottom of the window.
Report PDF shows a file path at the top or bottom of each page	Untick *Headers and footers* in the browser's print dialog; the report prints its own title and page numbers.
Profile Viewer opens the wrong file	Use Open... in the dock; layers from other sources are remembered per layer.

Symptom	Check
Complete Road Design stops	The GeoPackage already exists; choose a new folder.

A Criteria tables

Key values of the bundled criteria files. The files contain further tables (superelevation rates, critical grade lengths, run-off) and the source of each value.

Kenya RDM 1.3 (2025)

Design speed km/h	40	50	60	70	80	90	100	110	120
SSD, level (m) — T3.16	40	55	70	85	100	120	140	165	190
K crest — T6.1	5	10	17	30	45	67	100	140	185
K sag — T6.4	9	14	19	25	32	40	50	60	70
R min, e max 6 % (m) — T5.1	50	85	130	185	265	355	465	595	755
Barrier run-out (m) — T12.6	—	—	55	—	85	—	105	120	—

Maximum grade % (desirable / absolute) — T6.5	Flat	Rolling	Mountainous	Escarpment
Rural	3 / 5	4 / 6	6 / 8	7 / 9
Urban	3 / 5	6 / 8	8 / 10	8 / 10

Other values: eye height 1.05 m, object height 0.2 m (T3.13); minimum grade 0.5 %; combined grade 10 %; straights 6V to 20V m (4 km at 100 km/h and above); same-direction straights ≥ 200 m; constant grade of 60–100 m between crest and sag; barrier warrant 6 m fill steeper than 1:3; speed steps ≤ 20 km/h (10 preferred) with ≥ 1 km transitions.

AASHTO 2018 (metric)

Design speed km/h	40	50	60	70	80	90	100	110	120
SSD (m) — T3-1	50	65	85	105	130	160	185	220	250
K crest — T3-35	4	7	11	17	26	39	52	74	95
K sag — T3-37	9	13	18	23	30	38	45	55	63
Side friction f — T3-7	0.23	0.19	0.17	0.15	0.14	0.13	0.12	0.11	0.09

Other values: eye height 1.08 m, object height 0.6 m; minimum vertical curve length 0.6V m; minimum radius from $V^2/127(e + f)$; default e max 8 %; barrier warrant thresholds are configurable defaults (3 m fill steeper than 1:3) without run-out lengths.

LandXML Road & Terrain GIS Tools is free software released under the GNU General Public License version 2 or later, without warranty. Civil 3D is a trademark of Autodesk, Inc.; OpenRoads Designer is a trademark of Bentley Systems, Incorporated. The design criteria summarise published standards for convenience; the source documents govern.